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Tsinghua-Berkeley Shenzhen Institute

Learning From Data

Lecture 6: Backpropagation and Neural Networks

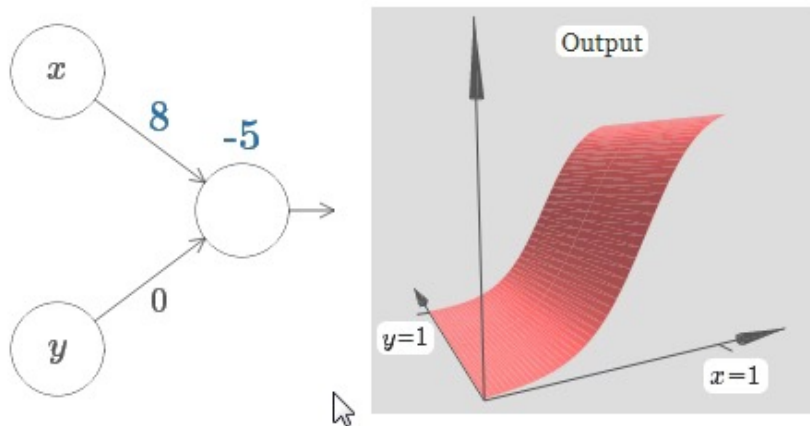
Yang Li

Slides by Lichen Wang

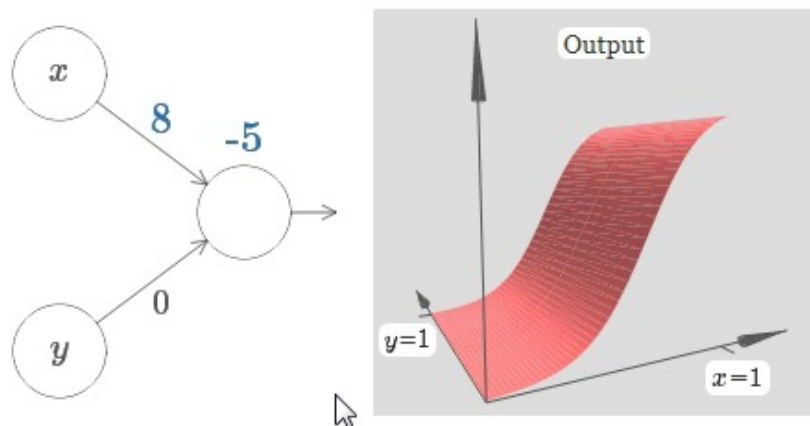
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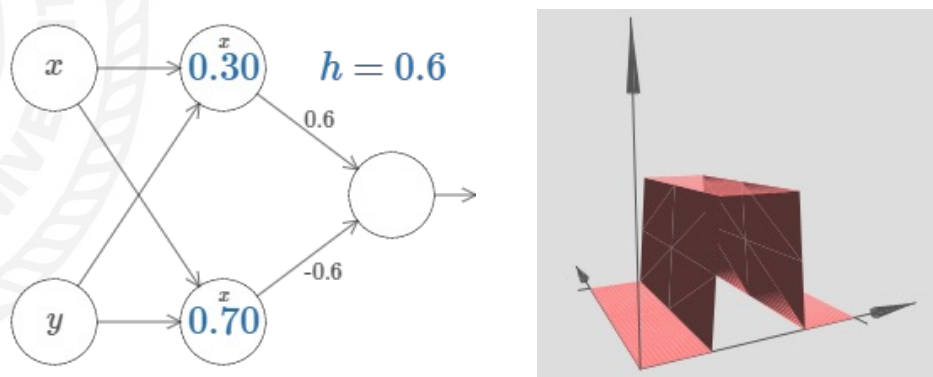
Power of single neural



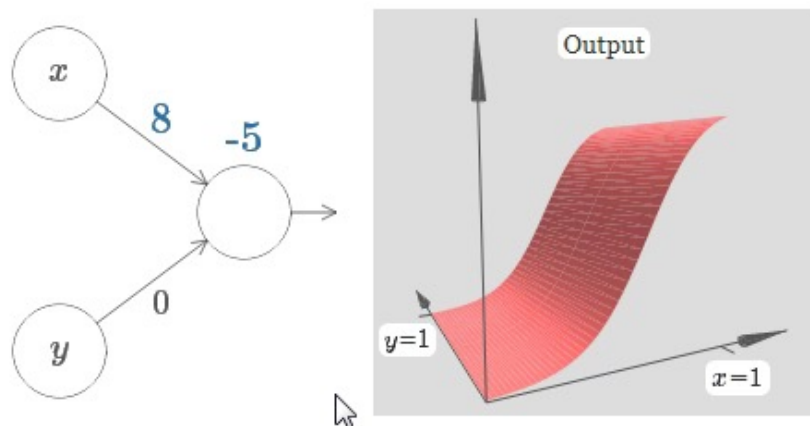
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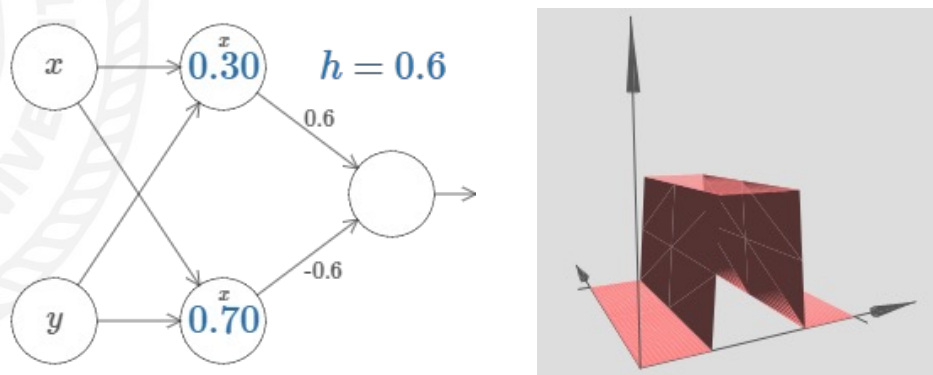
Two hidden units



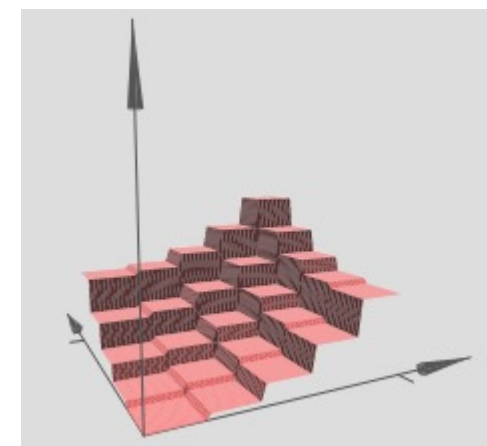
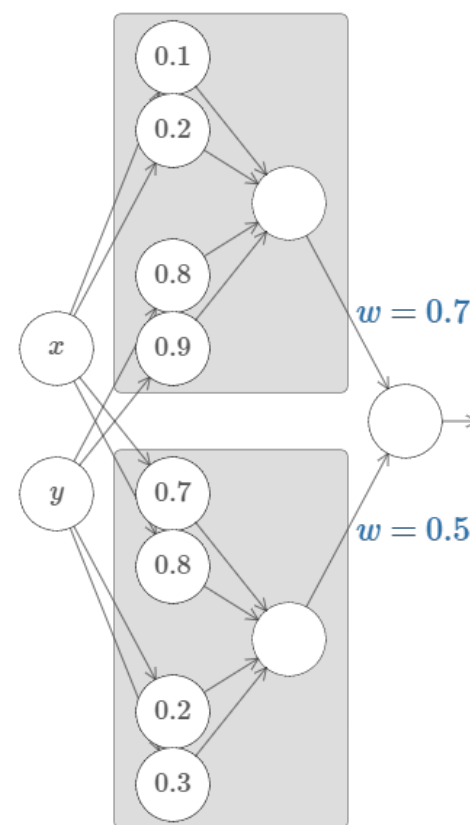
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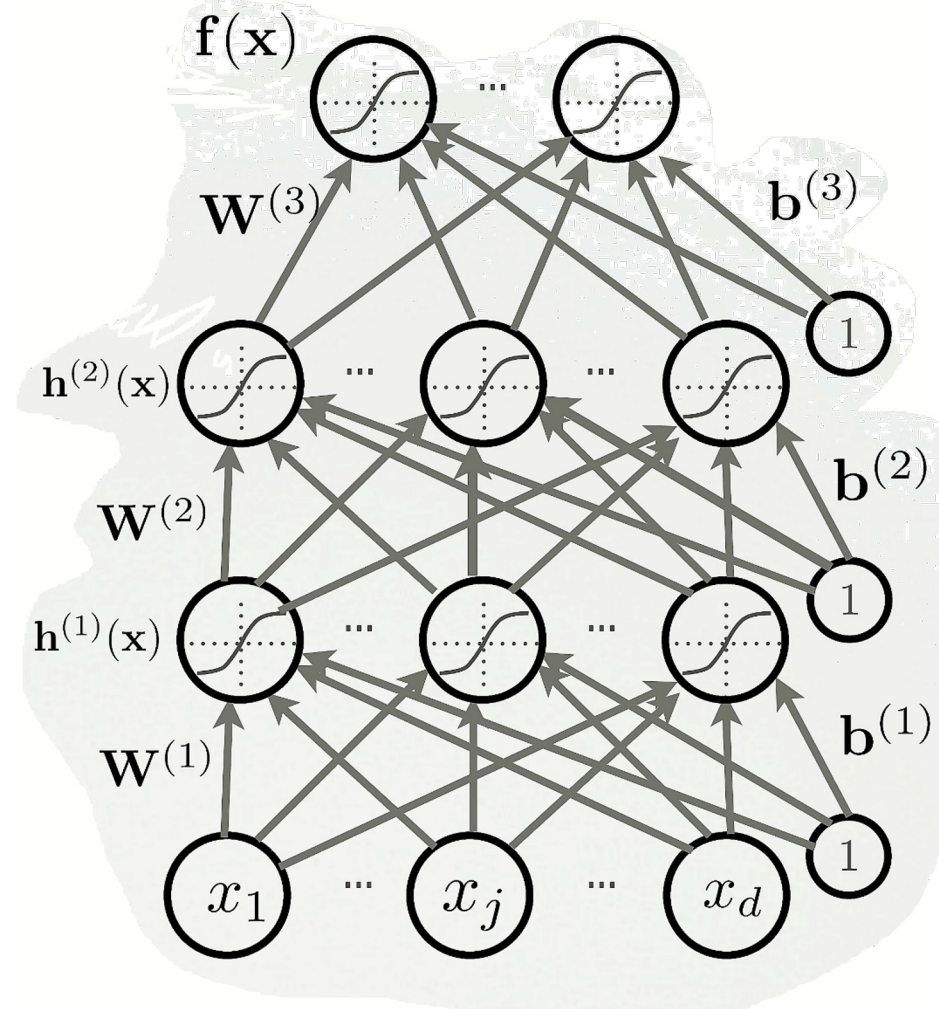
Many hidden units



Multilayer Neural Network



Could have L hidden layers

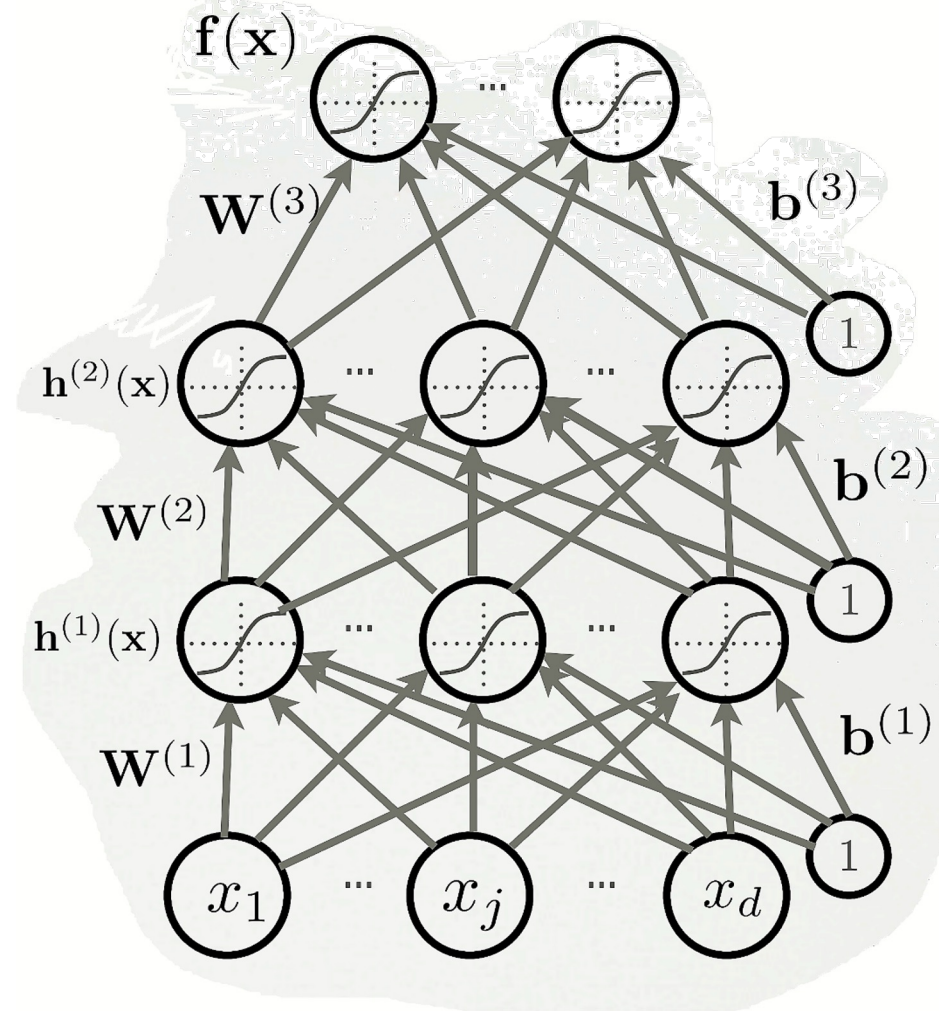


Multilayer Neural Network



Could have L hidden layers

- layer input activation for $k > 0$, $\mathbf{h}^{(0)}(\mathbf{x}) = \mathbf{x}$
 $\mathbf{a}^{(k)}(\mathbf{x}) = \mathbf{b}^{(k)} + \mathbf{W}^{(k)}\mathbf{h}^{(k-1)}(\mathbf{x})$



Multilayer Neural Network



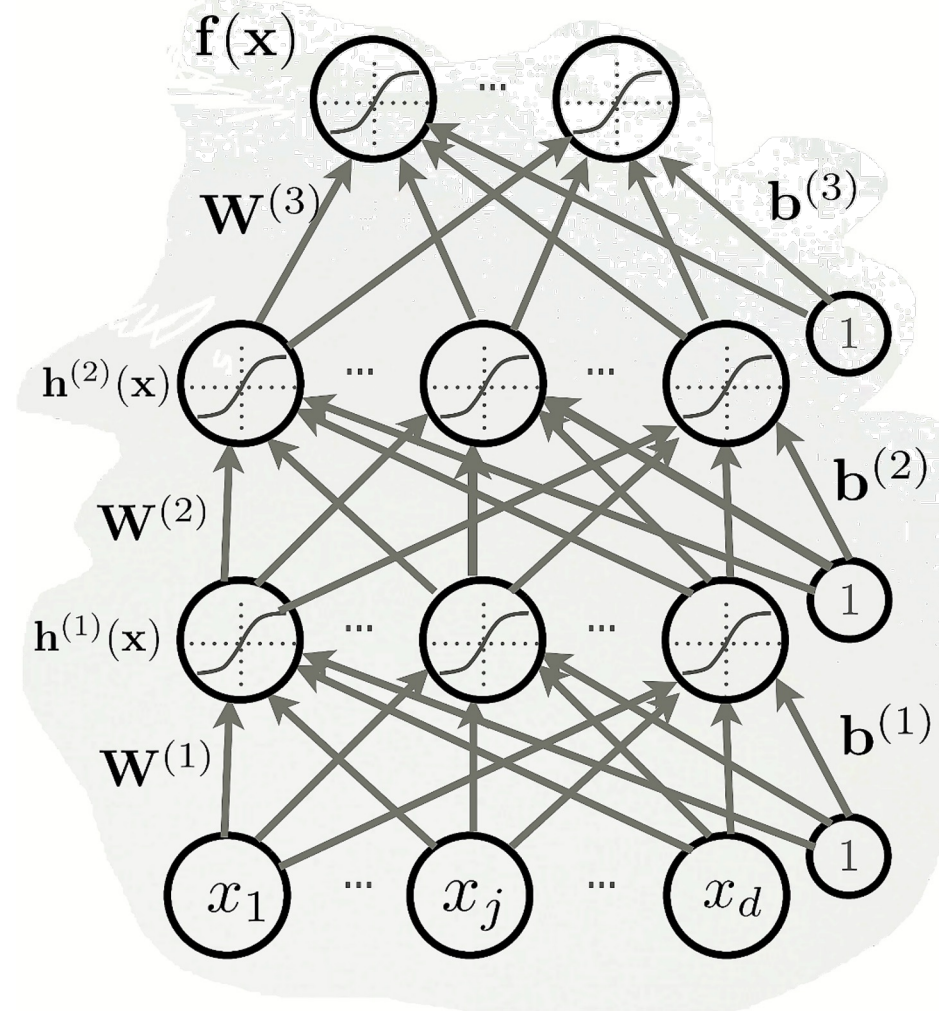
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- hidden layer activation for $1 \leq k \leq L$

$$\mathbf{h}^{(k)}(\mathbf{x}) = \mathbf{g}(\mathbf{a}^{(k)}(\mathbf{x}))$$



Multilayer Neural Network



Could have L hidden layers

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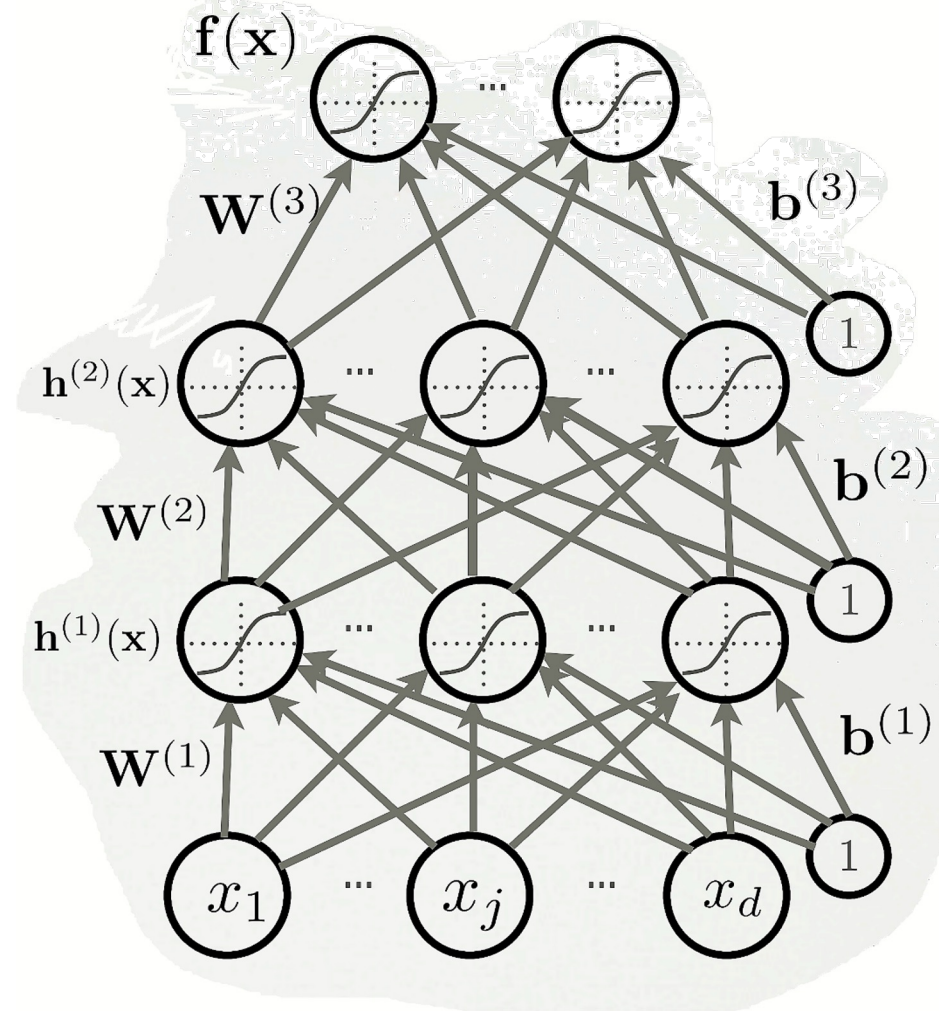
$$\mathbf{a}^{(k)}(\mathbf{x}) = \mathbf{b}^{(k)} + \mathbf{W}^{(k)} \mathbf{h}^{(k-1)}(\mathbf{x})$$

- hidden layer activation for $1 \leq k \leq L$

$$\mathbf{h}^{(k)}(\mathbf{x}) = \mathbf{g}(\mathbf{a}^{(k)}(\mathbf{x}))$$

- output layer activation for $k = L + 1$

$$\mathbf{h}^{(L+1)}(\mathbf{x}) = \mathbf{g}(\mathbf{a}^{(L+1)}(\mathbf{x})) = \mathbf{f}(\mathbf{x})$$



Empirical risk

$$\operatorname{argmin}_W \frac{1}{m} \sum_{i=1}^m L(f(\mathbf{x}^{(i)}; W), y^{(i)}) + \lambda \Omega(W)$$

Penalized version also known as
Structured Risk Minimization

■ $L(f(\mathbf{x}^{(i)}; W), y^{(i)})$ is the loss function for sample $(\mathbf{x}^{(i)}, y^{(i)})$

■ $\lambda \Omega(W)$ is the regularizer

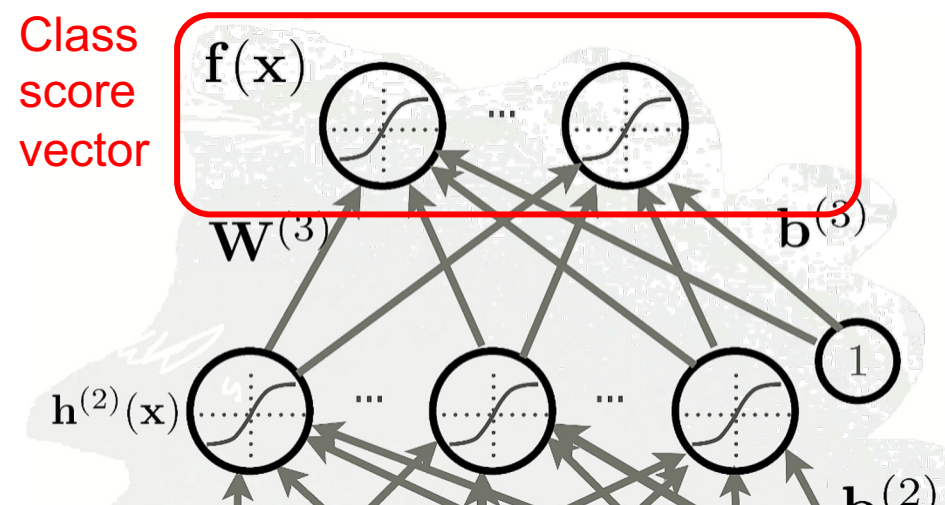
e. g. When L is cross entropy loss (log softmax+NLL)

$$L(f(\mathbf{x}^{(i)}; W), y^{(i)}) = -\log \left(\frac{e^{f_{y^{(i)}}}}{\sum_{j=1}^{|Y|} e^{f_j}} \right)$$

f_j is the j th element of class score vector $f(\mathbf{x}^{(i)}; W)$

Unnormalized class probability of $|Y|$ classes

Class
score
vector





- Find the optimal parameter

$$\operatorname{argmin}_W \frac{1}{m} \sum_{i=1}^m L(f(\mathbf{x}^{(i)}; W), y^{(i)}) + \lambda \Omega(W)$$

■ Find the optimal parameter

$$\operatorname{argmin}_W \frac{1}{m} \sum_{i=1}^m L(f(\mathbf{x}^{(i)}; W), y^{(i)}) + \lambda \Omega(W)$$

■ Stochastic Gradient Descent (SGD)

Algorithm

1. Initialize $\mathbf{W}$

repeat: for each training example $(\mathbf{x}^{(t)}, y^{(t)})$

2a. $\Delta = -\nabla_{\mathbf{w}} L(f(\mathbf{x}^{(t)}; \mathbf{W}), y^{(t)}) - \lambda \nabla_{\mathbf{w}} \Omega(\mathbf{W})$

2b. $\mathbf{W} \leftarrow \mathbf{W} + \alpha \Delta$

Training epoch
=
Iterating over all examples

To apply this algorithm, we need:

1. A procedure to compute the parameter gradient
2. The regularizer (and its gradient)
3. Updating rule
4. Initialization method

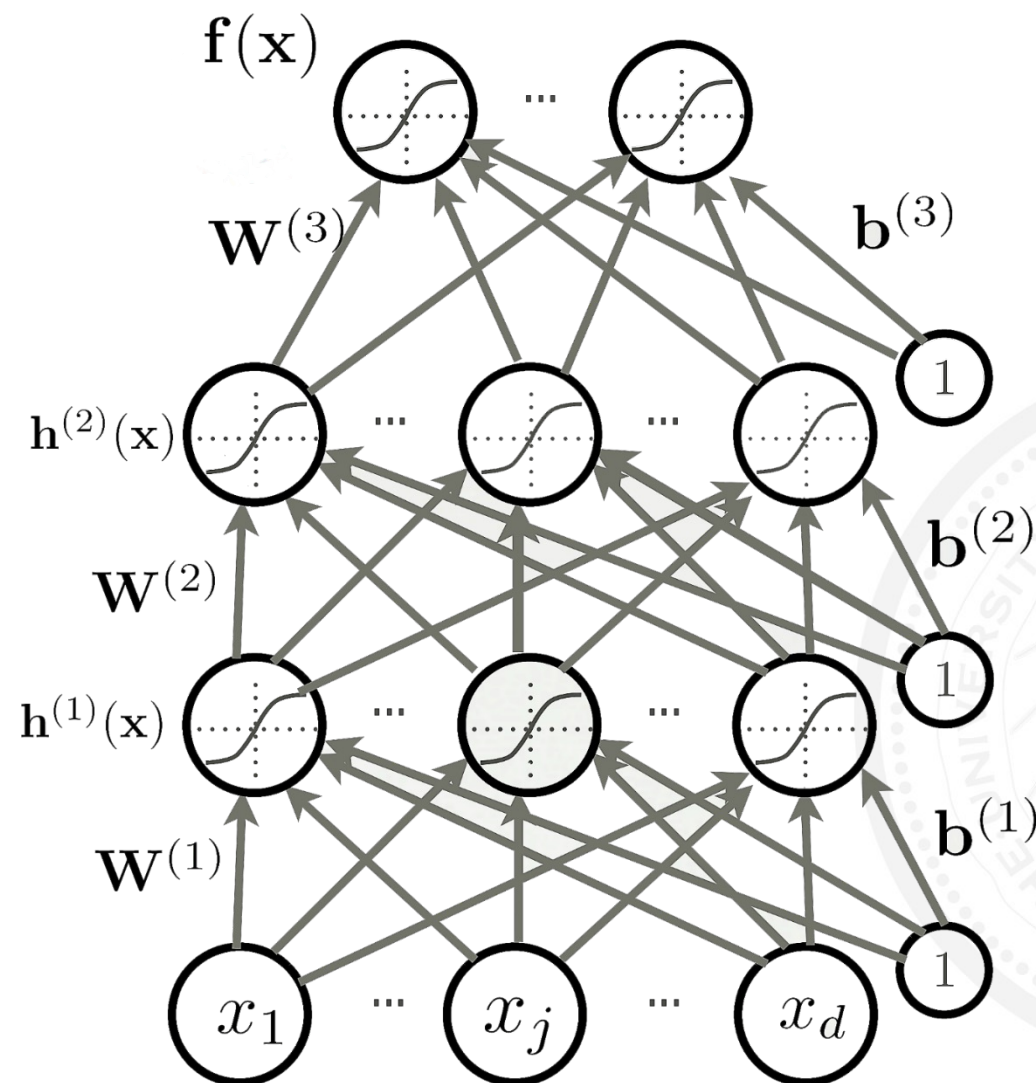
Animation:

<http://vision.stanford.edu/teaching/cs231n-demos/linear-classify/>

Follow the slope



How many parameter do we have?

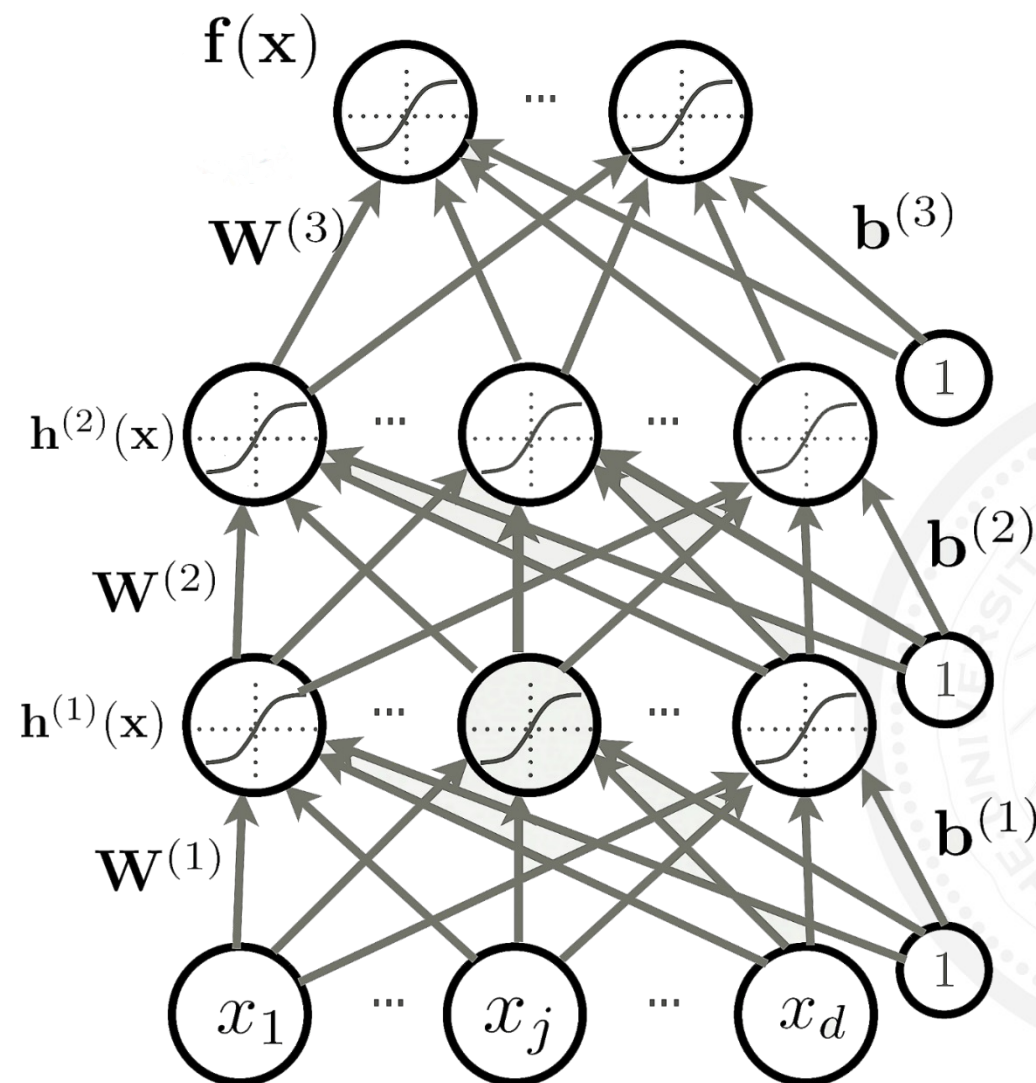


Follow the slope



How many parameter do we have?

VGGNet [Simonyan and Zisserman, 2014] used 138M parameters



Follow the slope

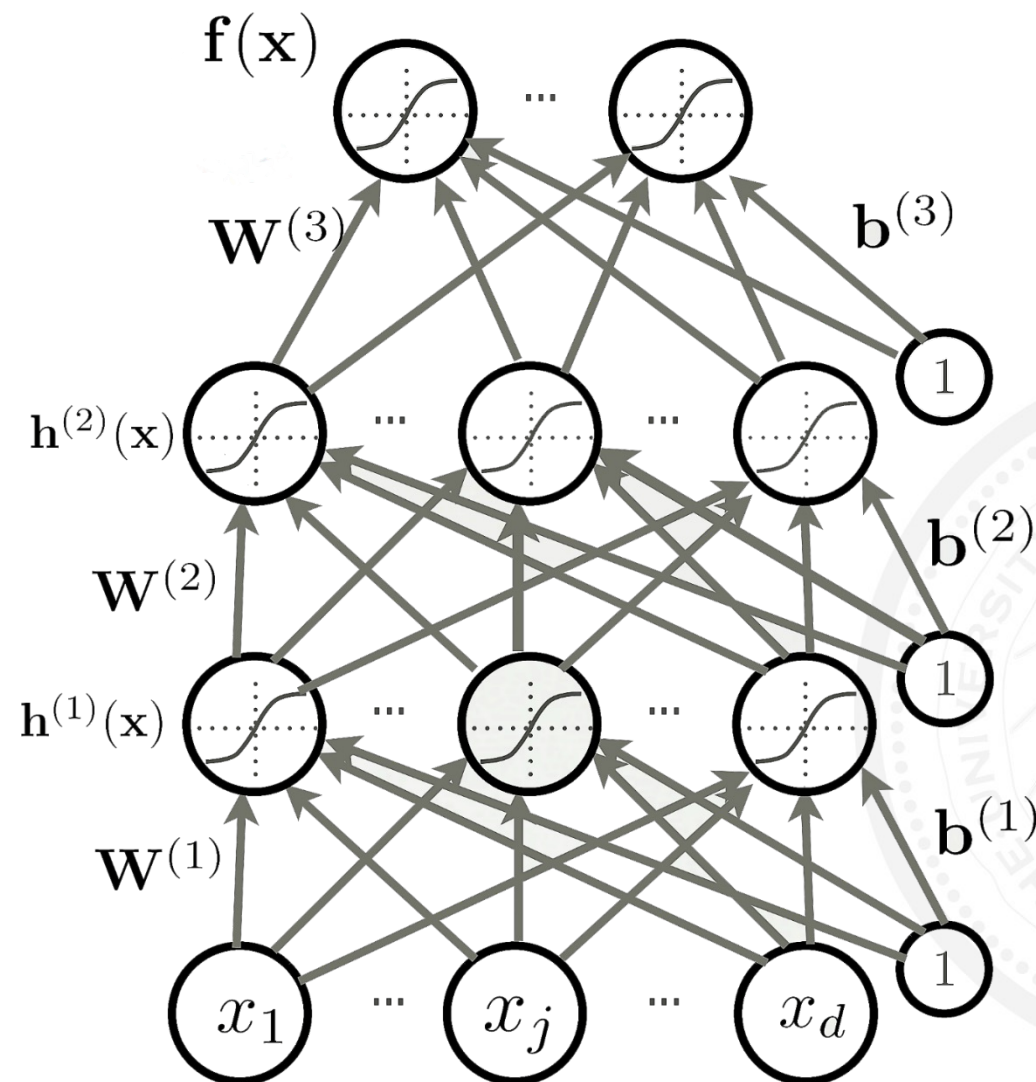


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In 1-dimension, the derivative of a function:

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



Follow the slope

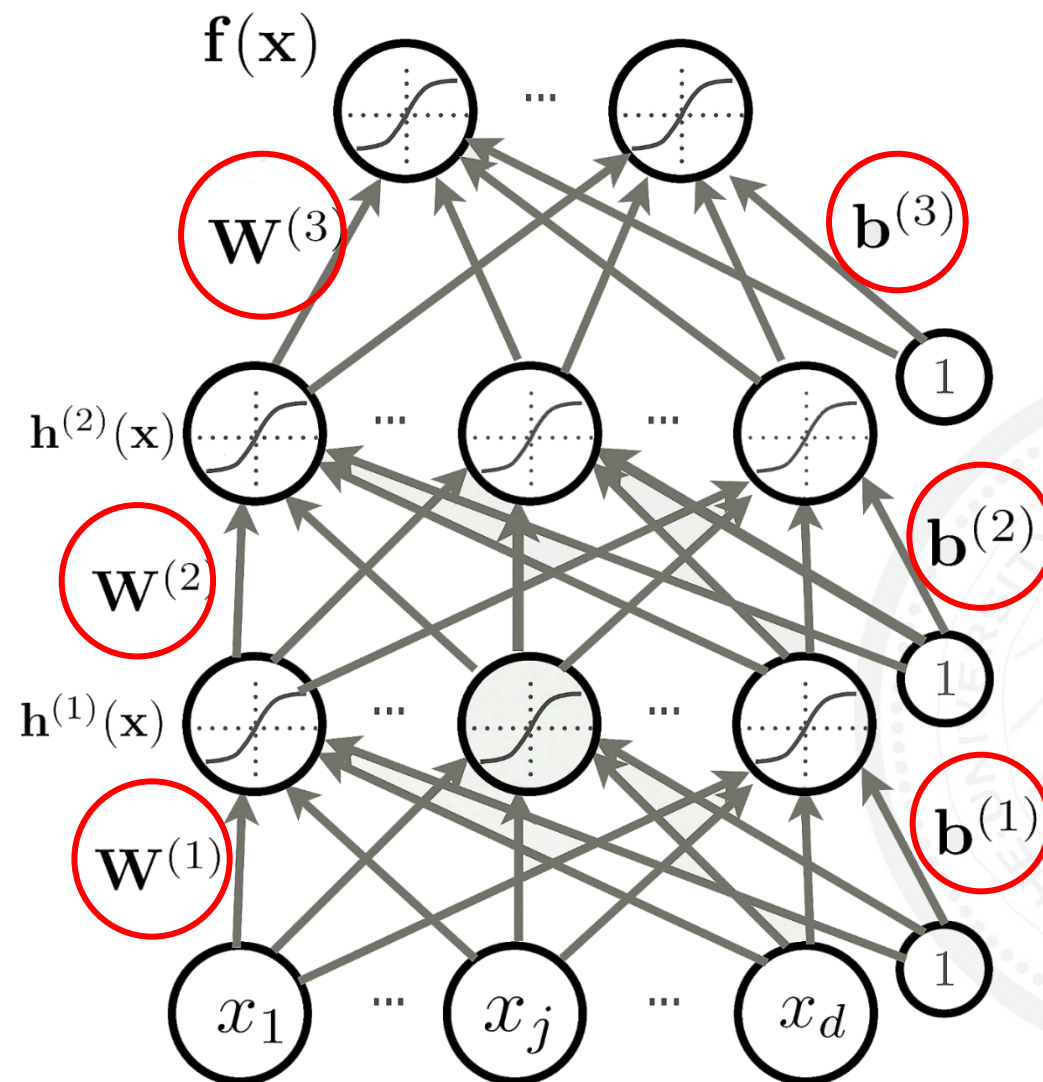


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Current W :

Gradient dW :





Current W:

[0.25,
-1.56,
0.55,
3.8,
0.98,
0.77,
-0.11,
-2.9,...]

Loss 1.25742

Gradient dW:

[? ,
?,
?,
?,
?,
?,
?,
?,...]



Current W:

[0.25,
-1.56,
0.55,
3.8,
0.98,
0.77,
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Loss 1.25742

W + h (third dim):

[0.25 + 0.0001,
-1.56,
0.55,
3.8,
0.98,
0.77,
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Loss 1.25763

Gradient dW:

[? ,
?,
?,
?,
?,
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?,
?,...]



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Loss 1.25742

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[0.25 + 0.0001,
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0.98,
0.77,
-0.11,
-2.9,...]

Loss 1.25763

Gradient dW:

[2.1,
?,
?,
?,
?,
?,
?,
?,...]

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{(1.25763 - 1.25742)}{0.0001}$$

Follow the slope

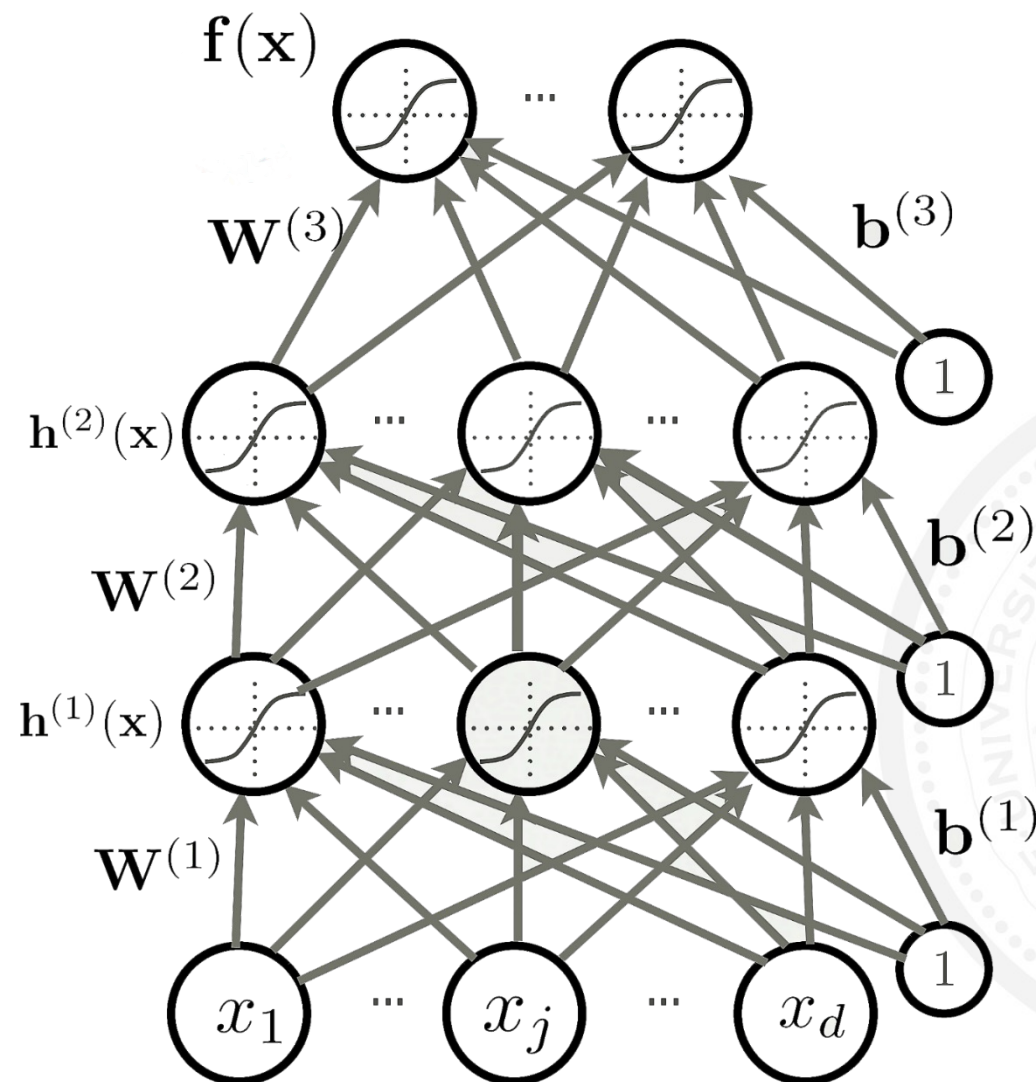


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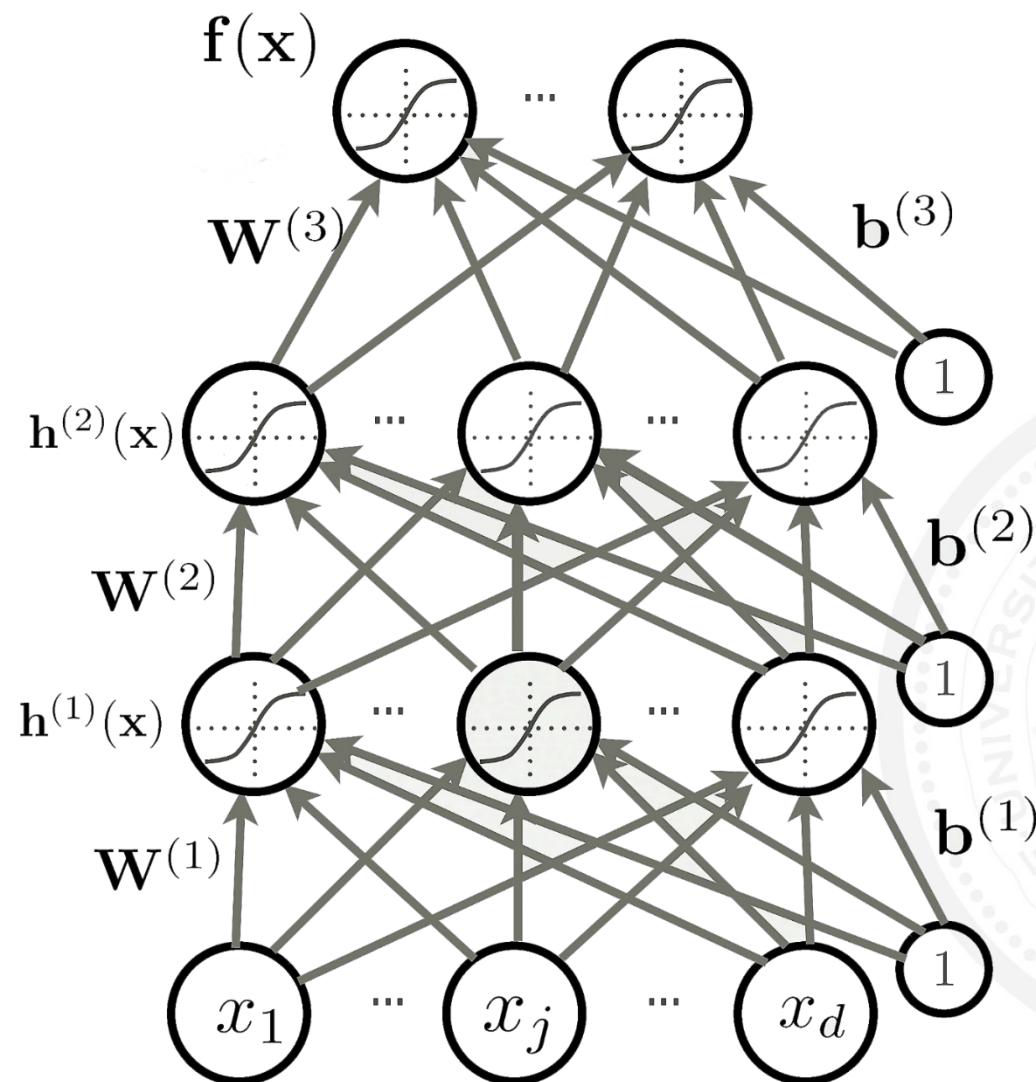
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Numerical gradient: approximate, slow, easy to write



Follow the slope



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Calculus!

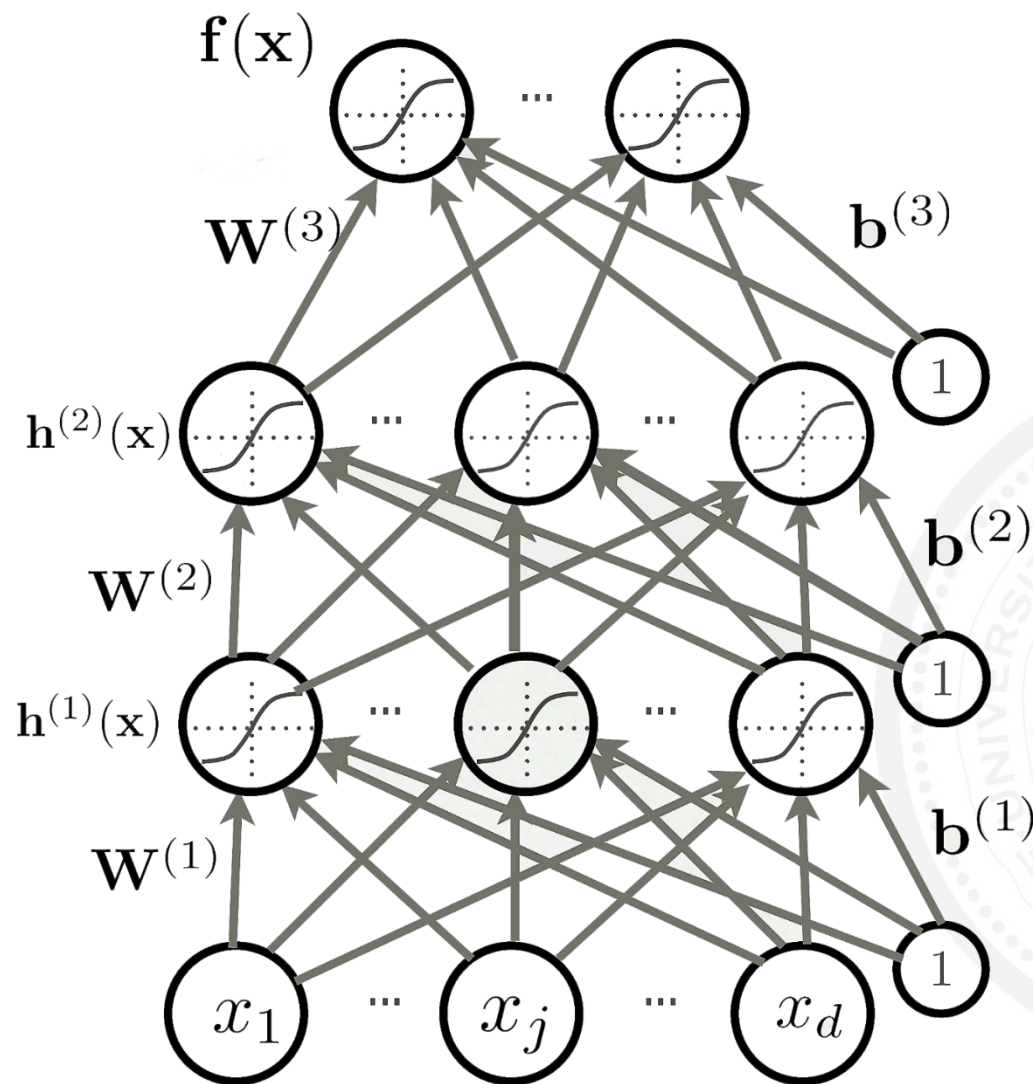
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Analytic gradient: exact, fast, error-prone

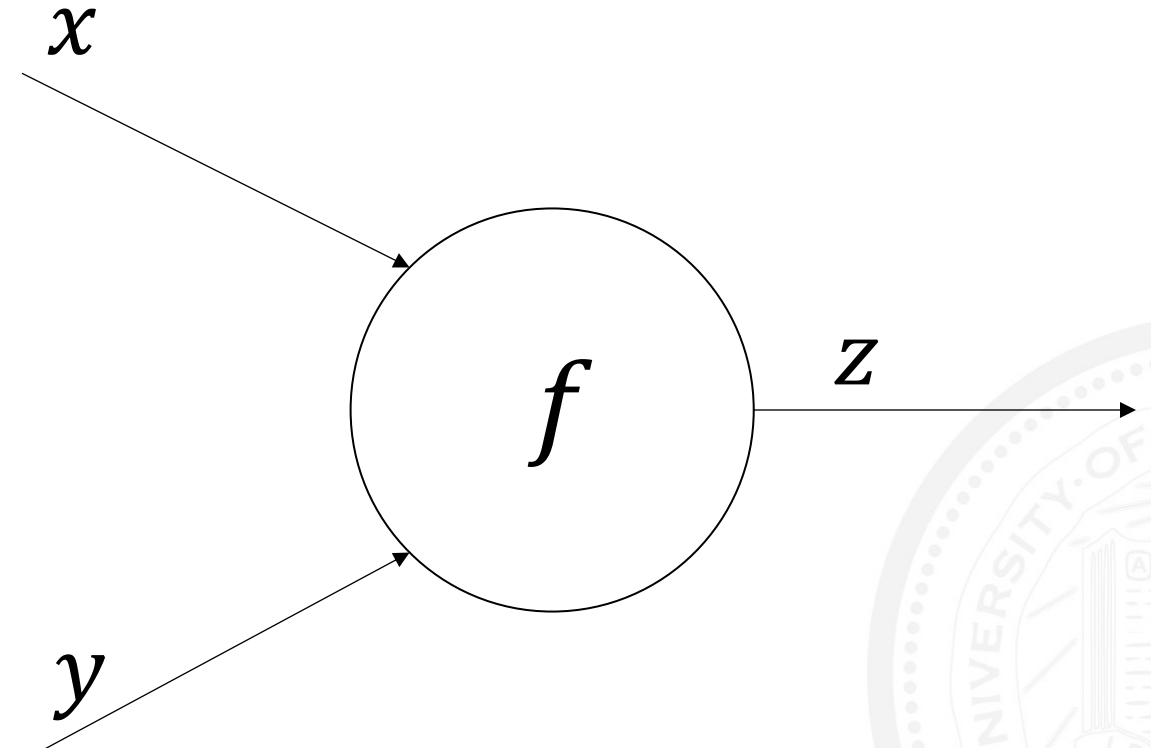
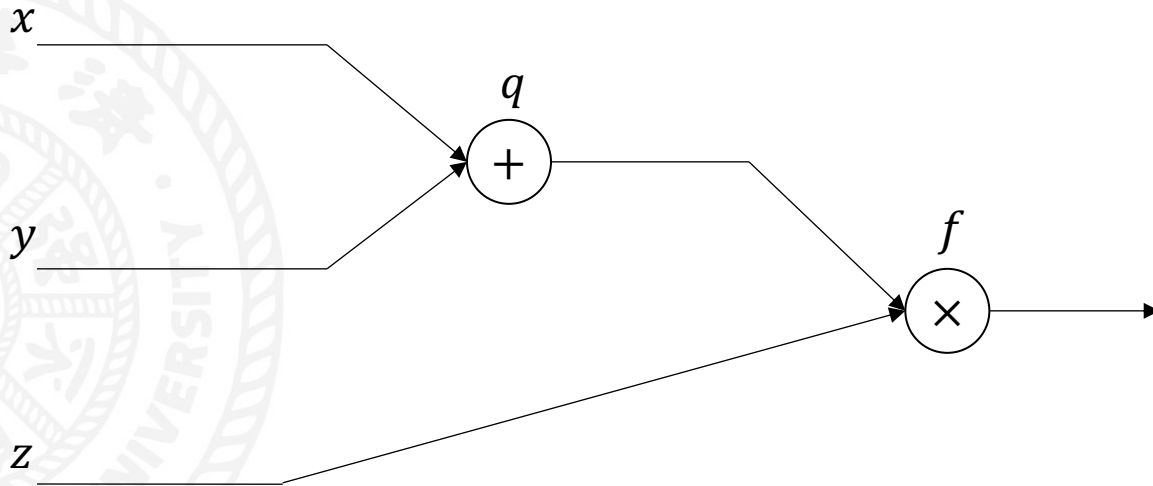


Backpropagation



$$f(x, y, z) = (x + y)z$$

We want $\frac{df}{dx}, \frac{df}{dy}, \frac{df}{dz}$



Backpropagation

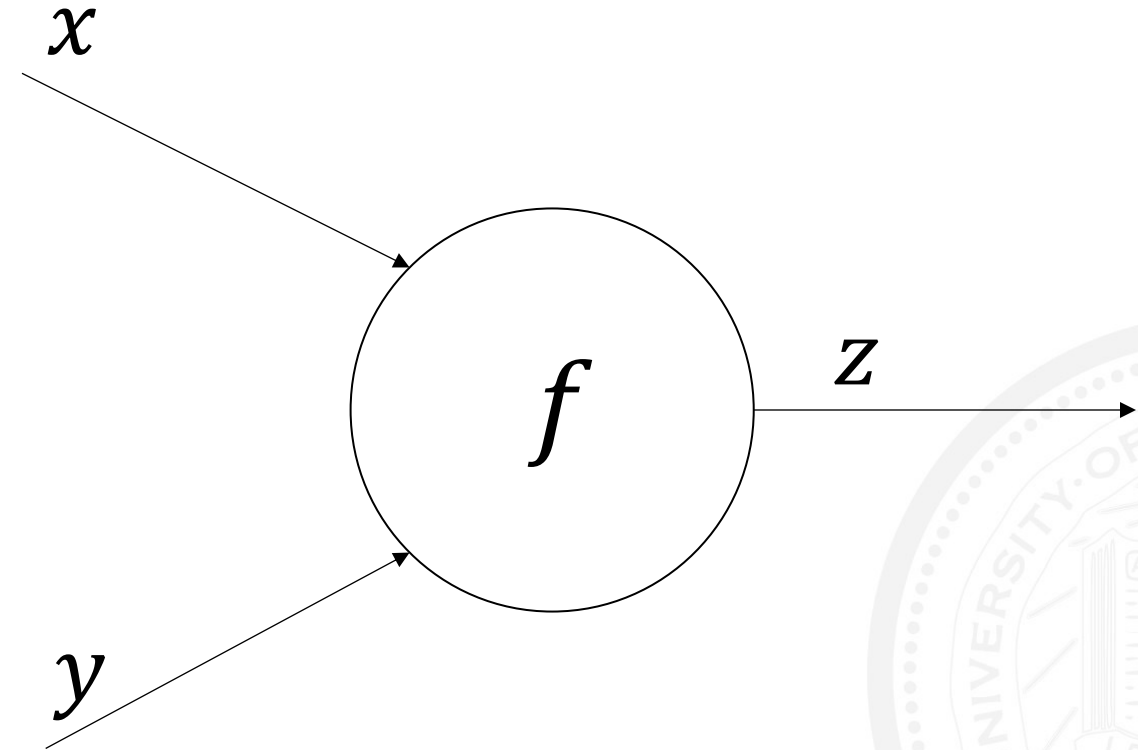
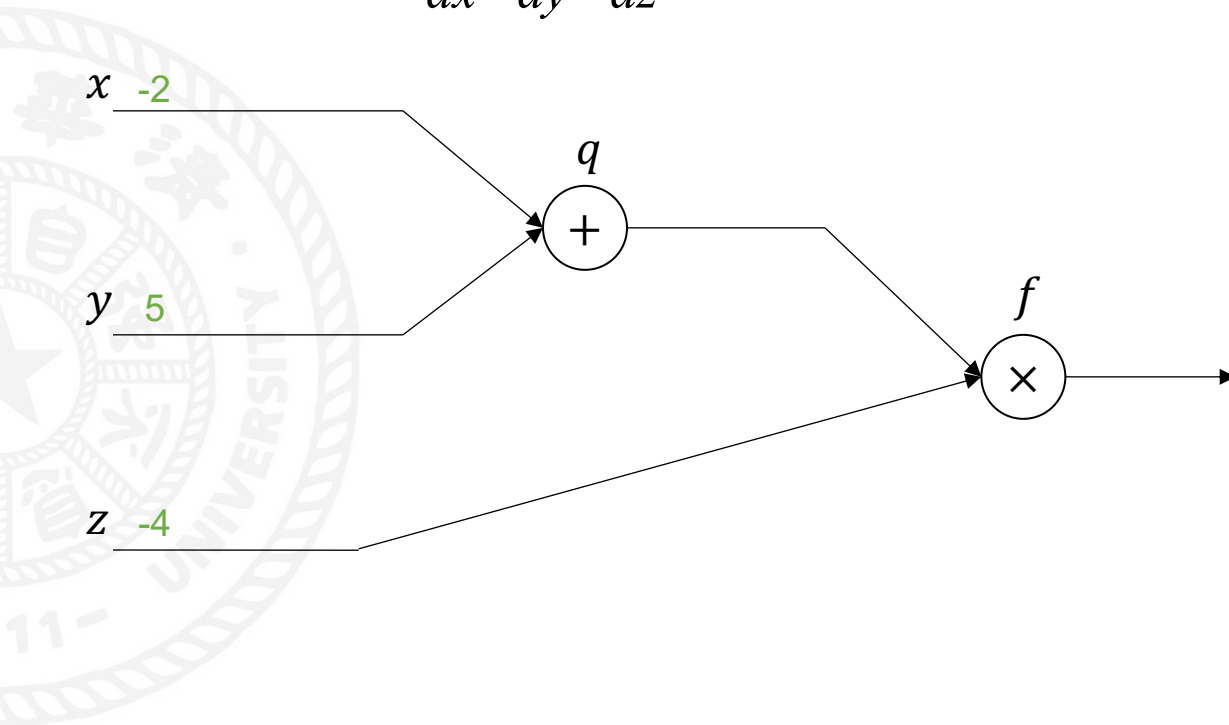


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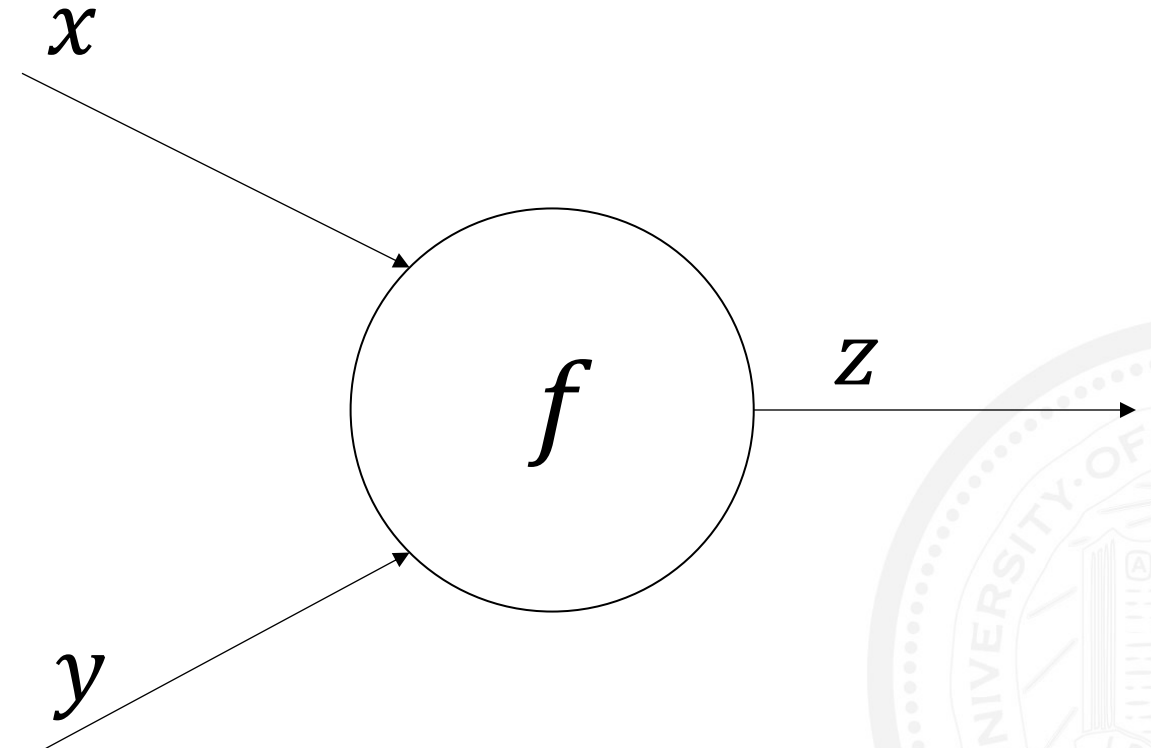
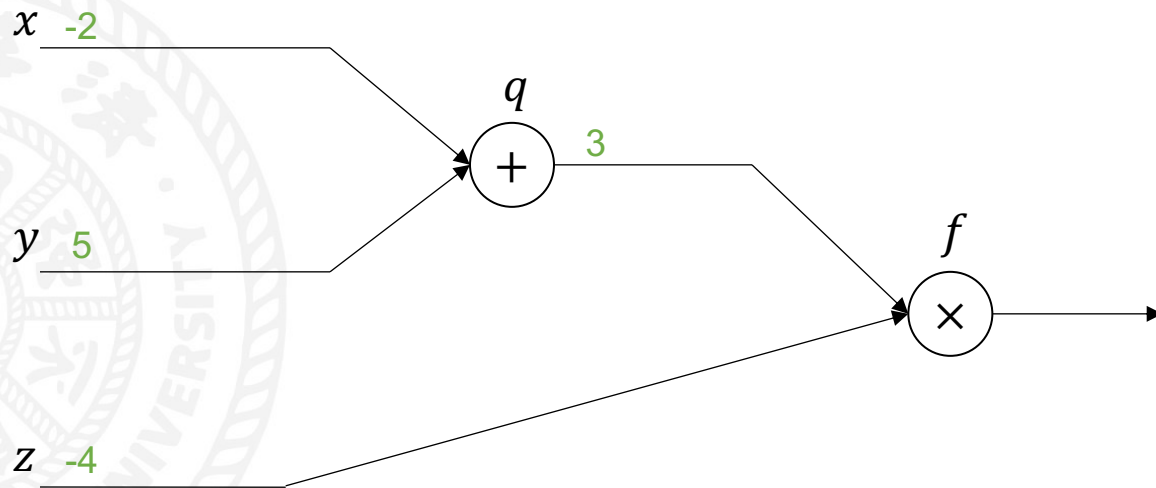


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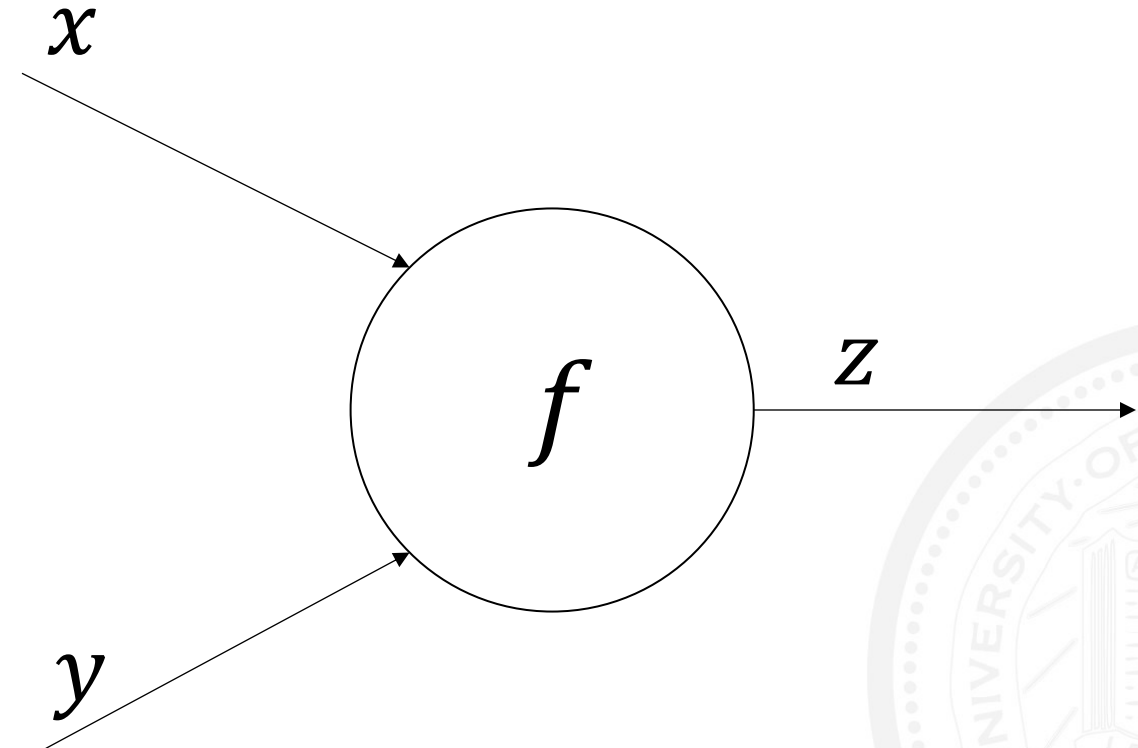
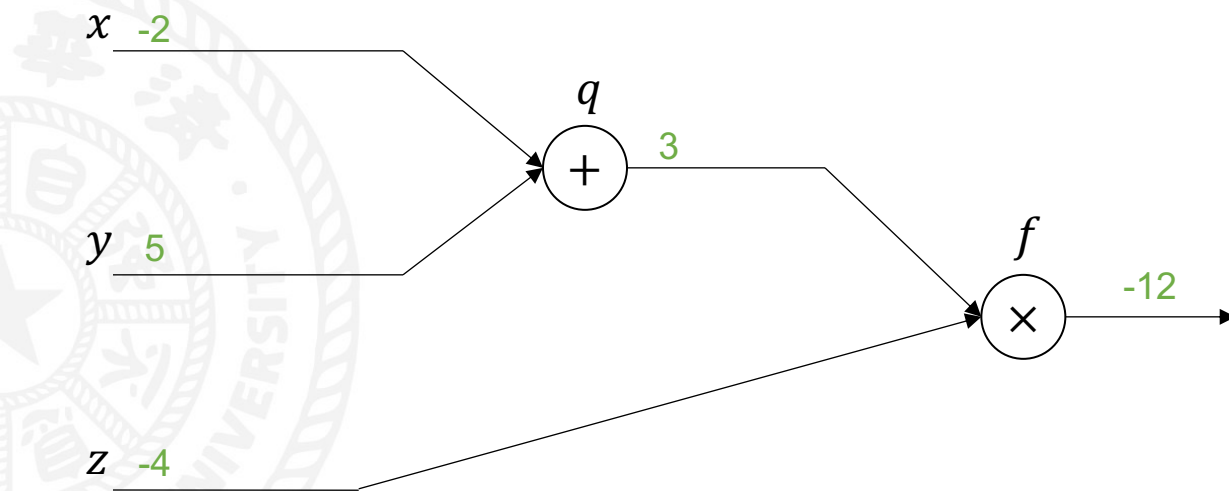


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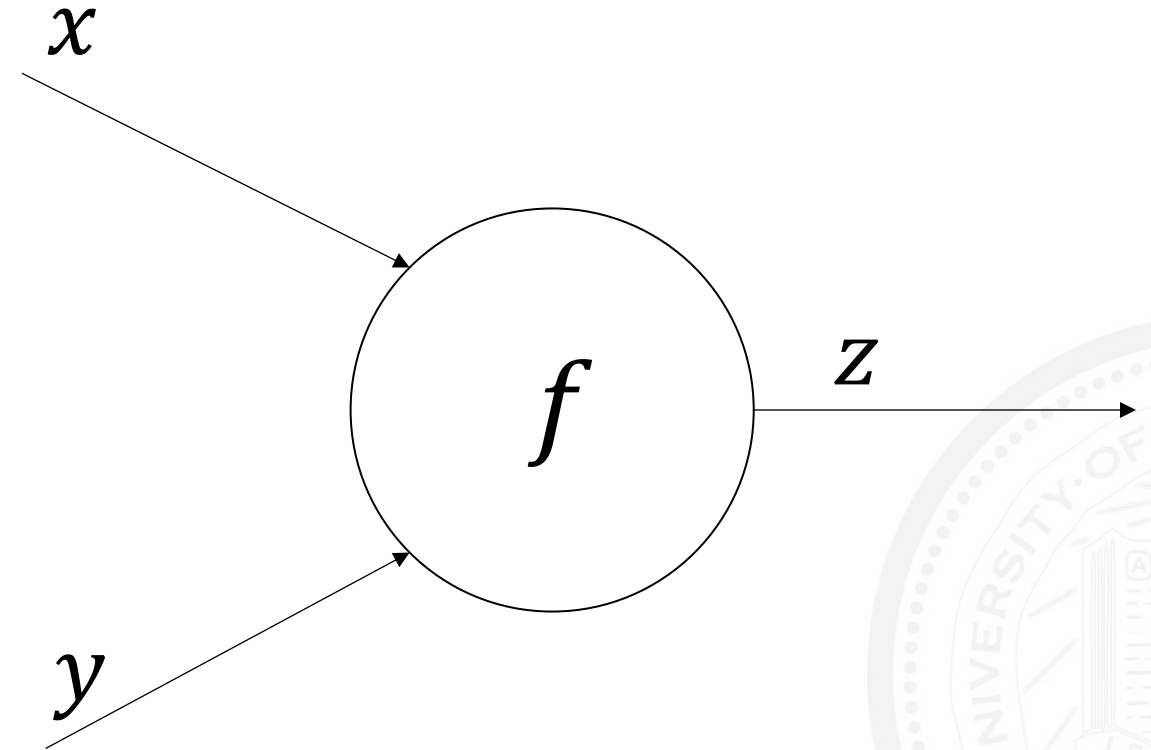
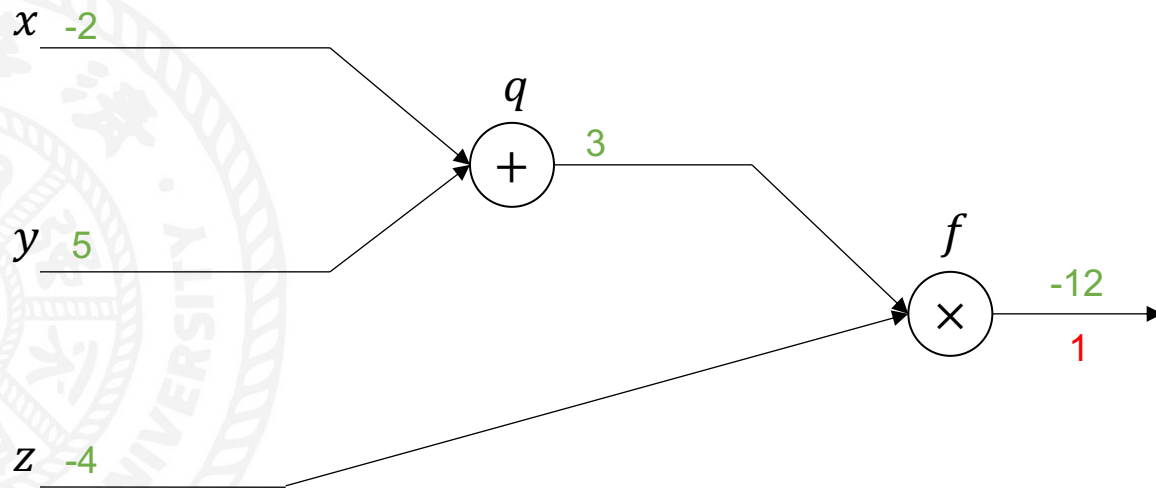


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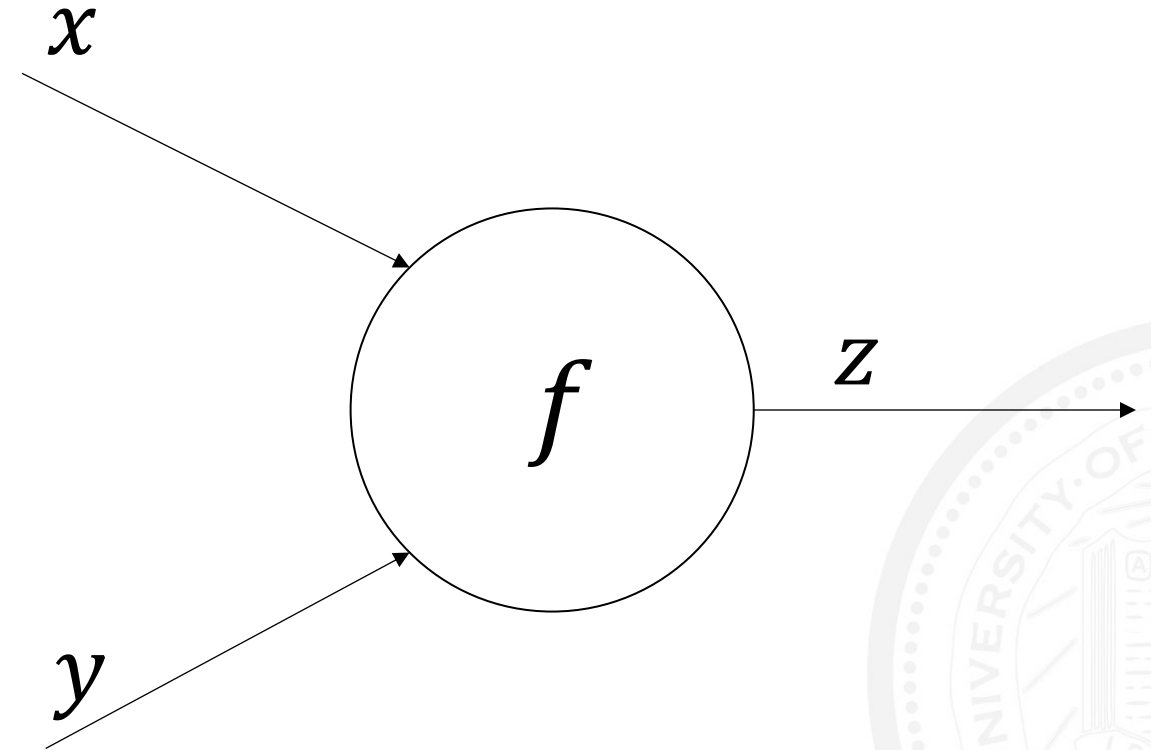
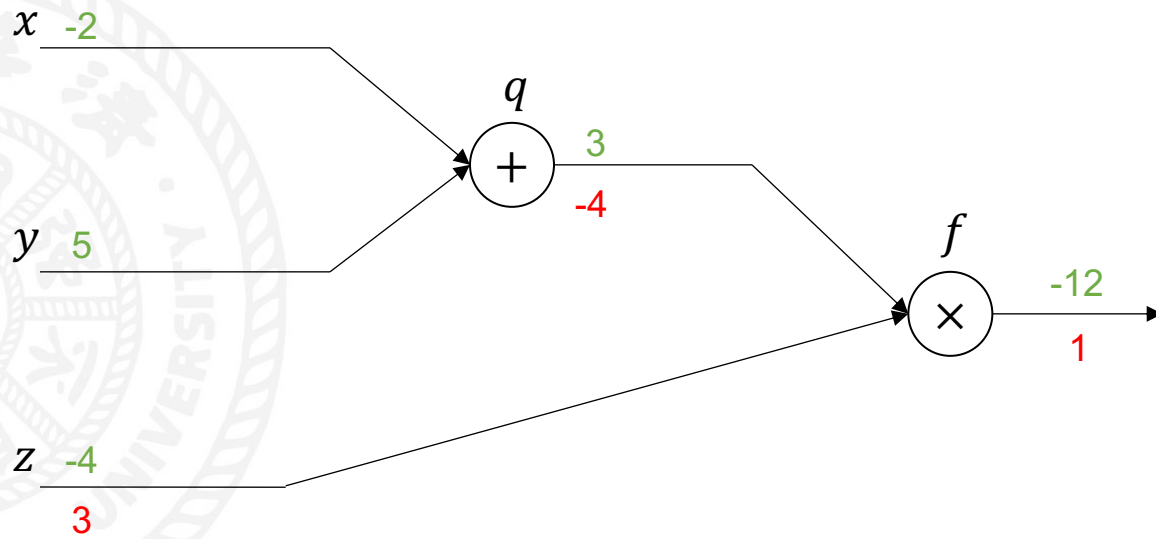


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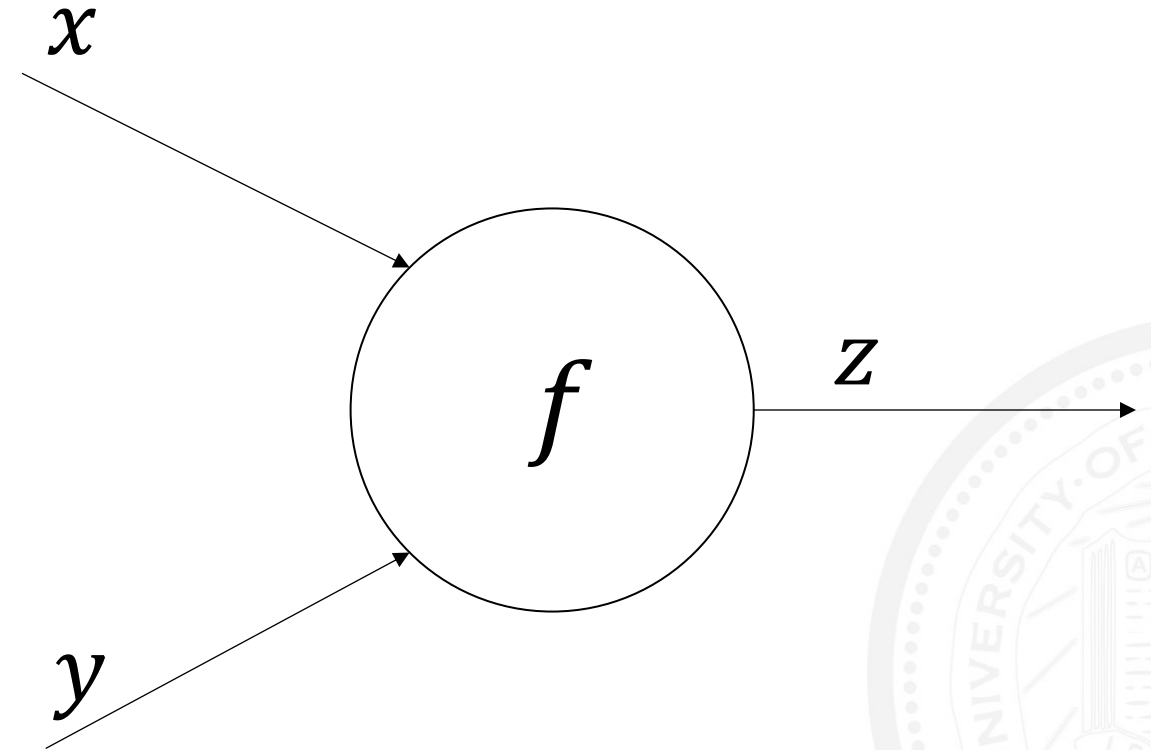
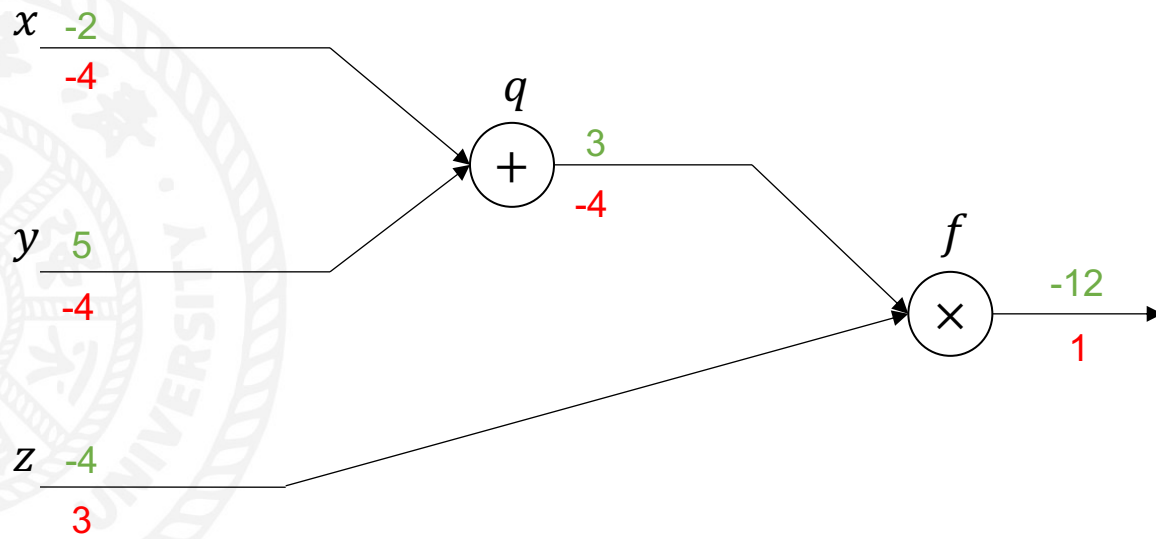


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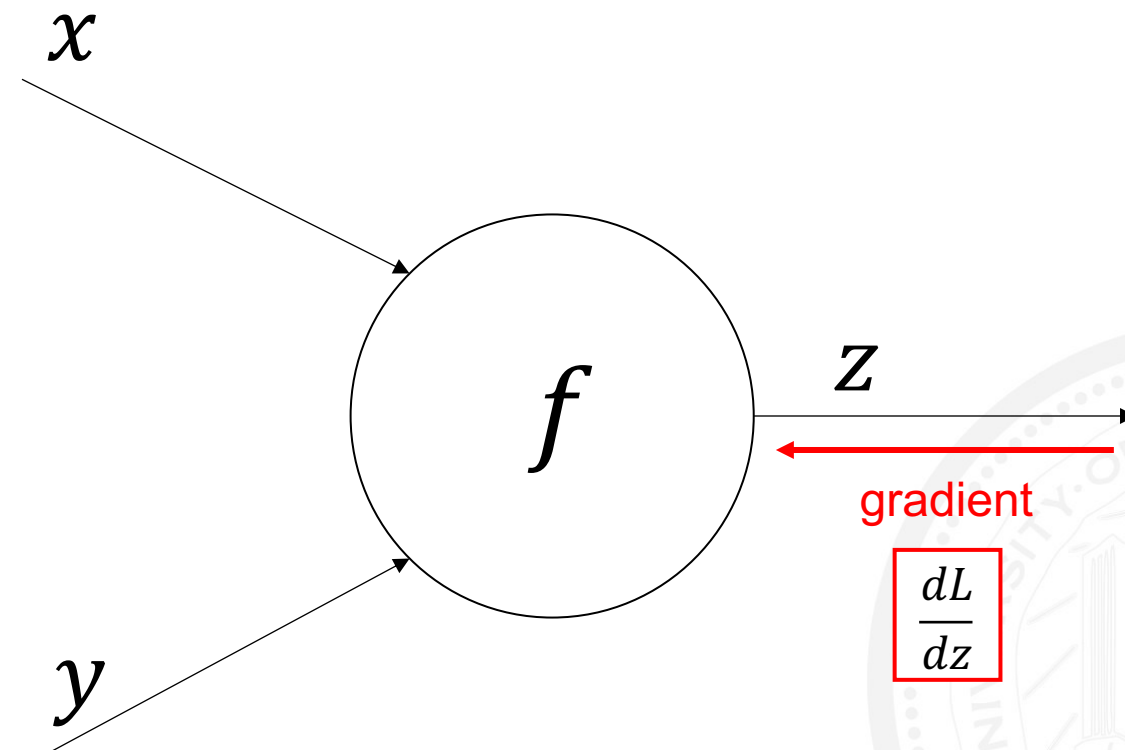
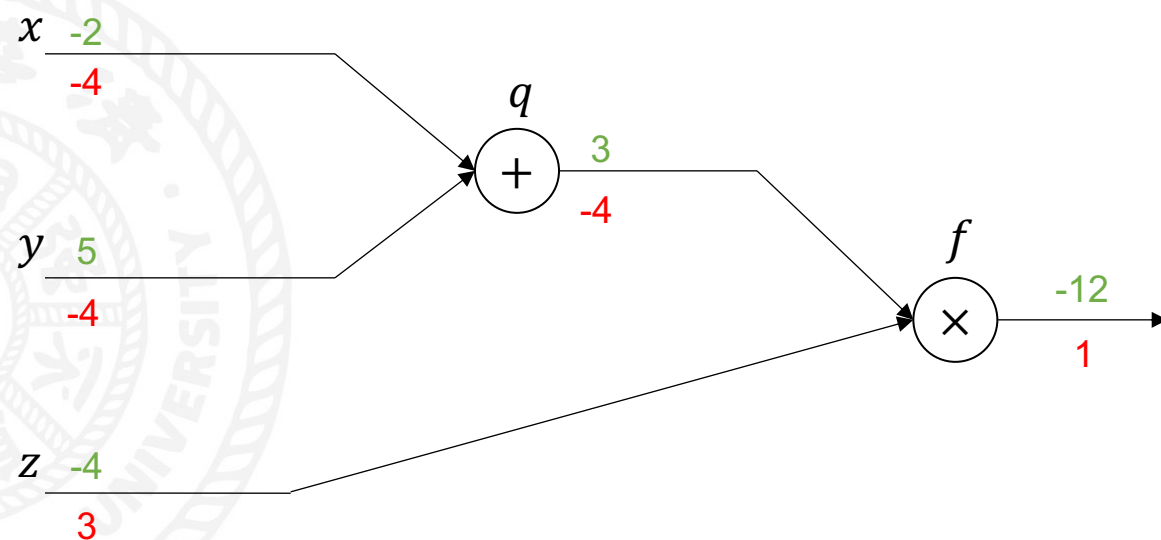


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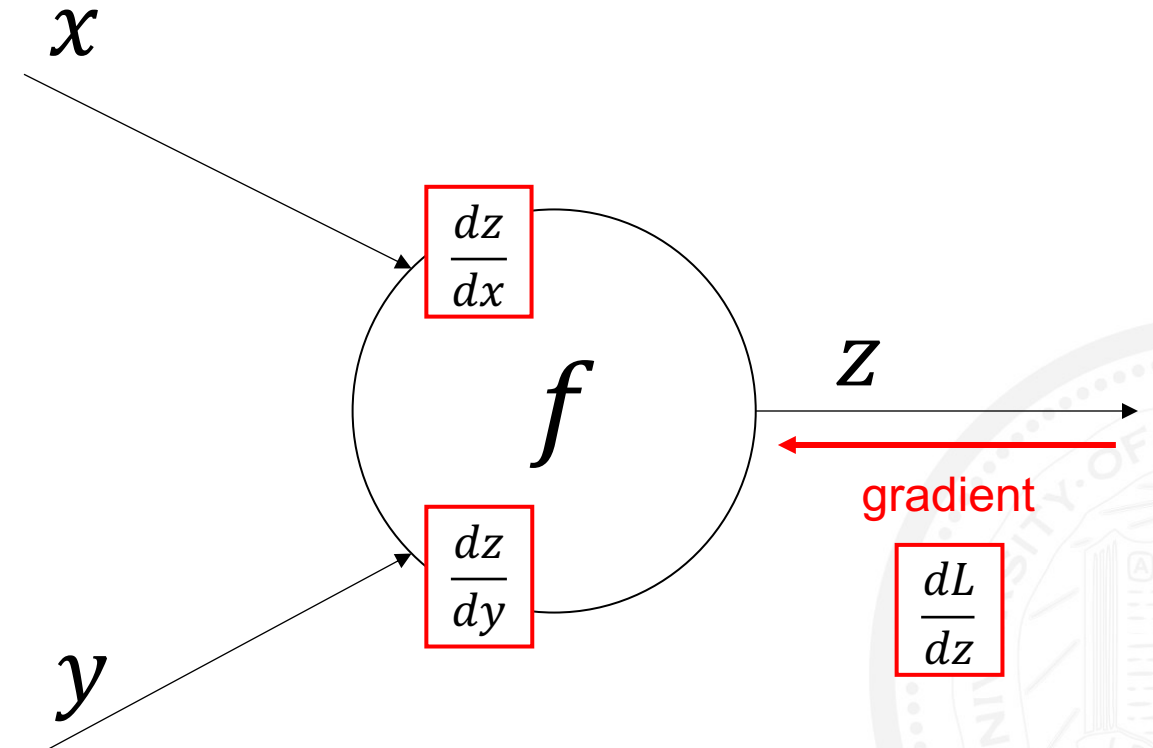
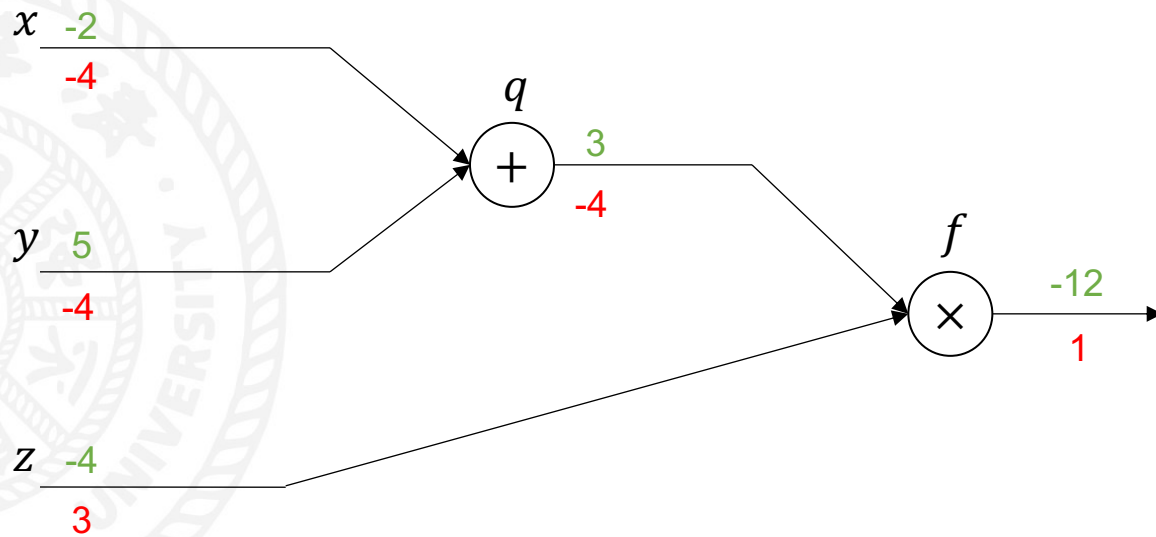


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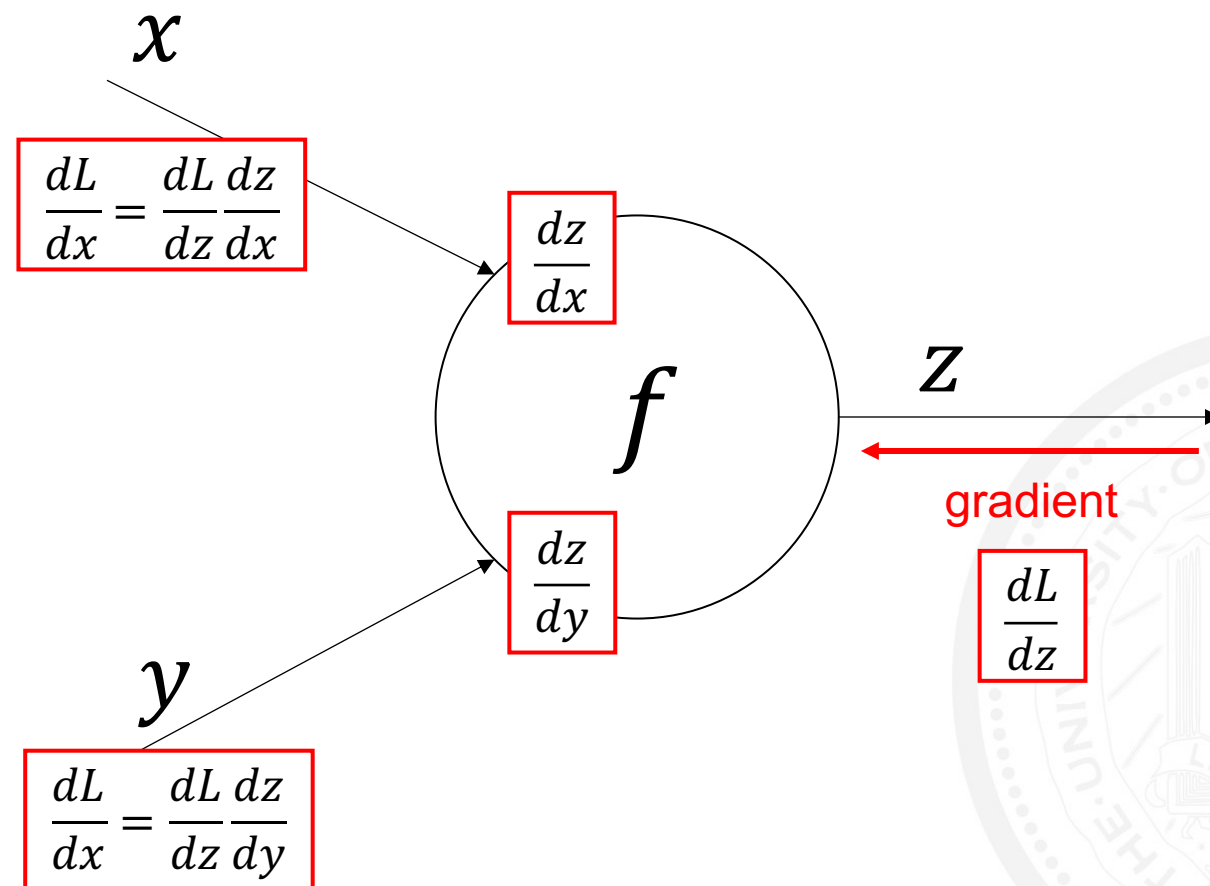
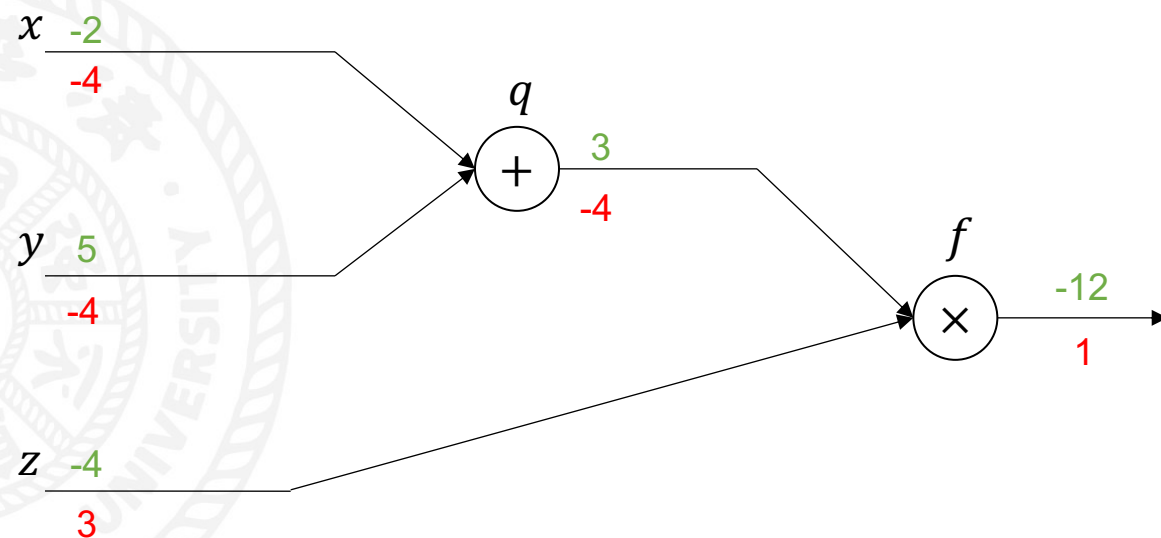


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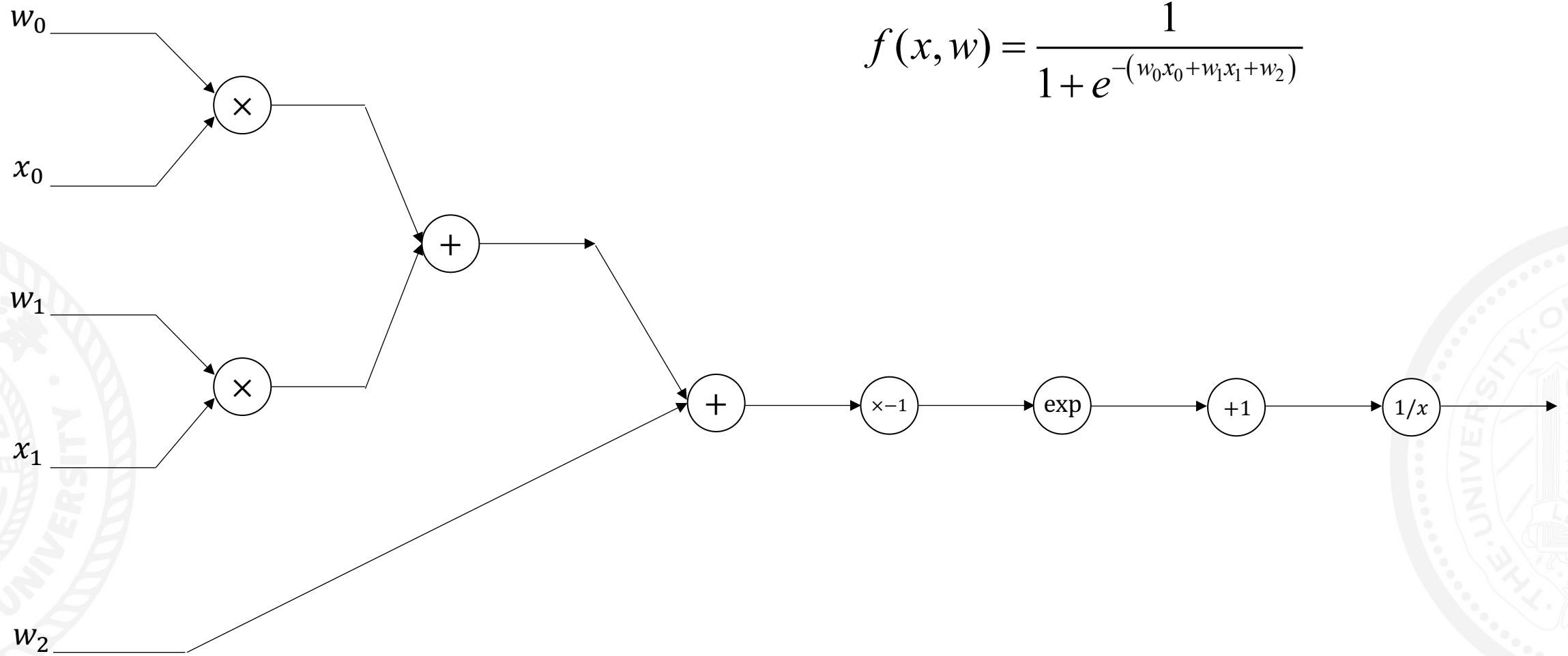


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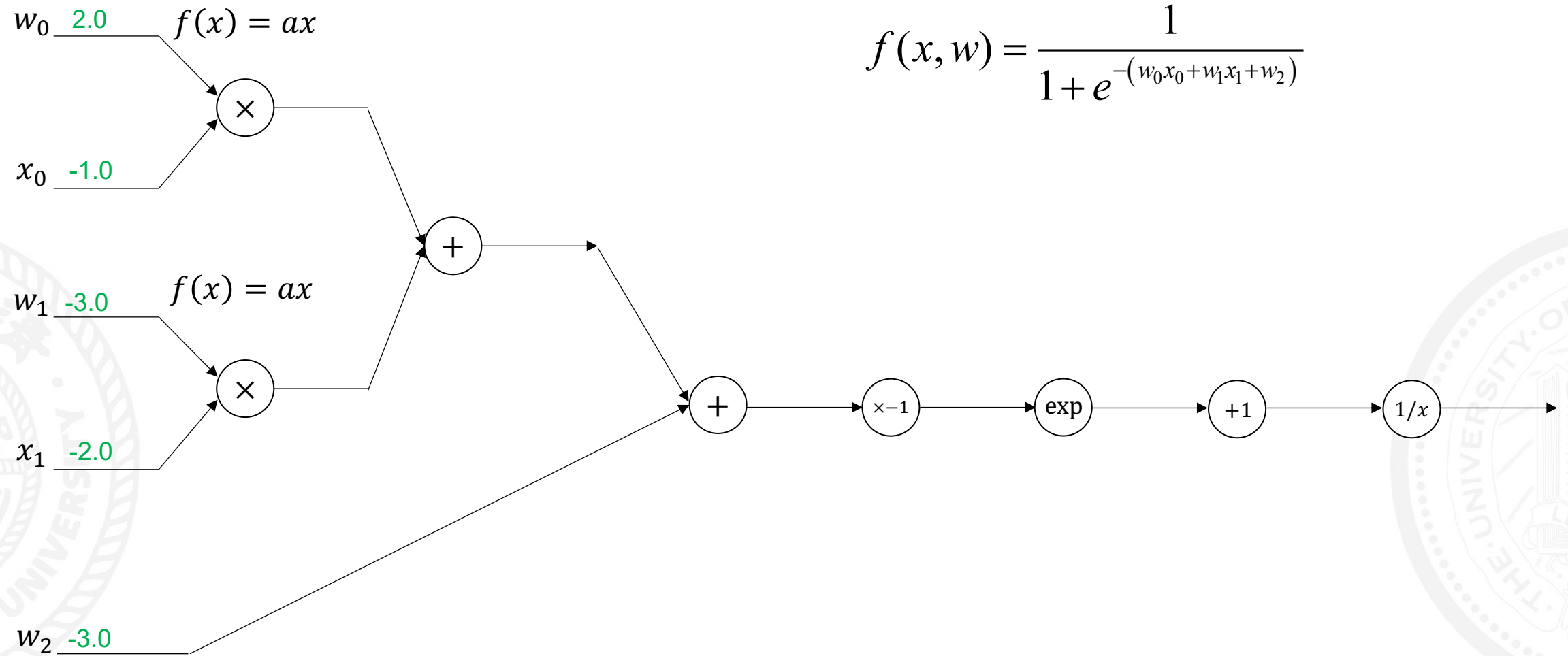


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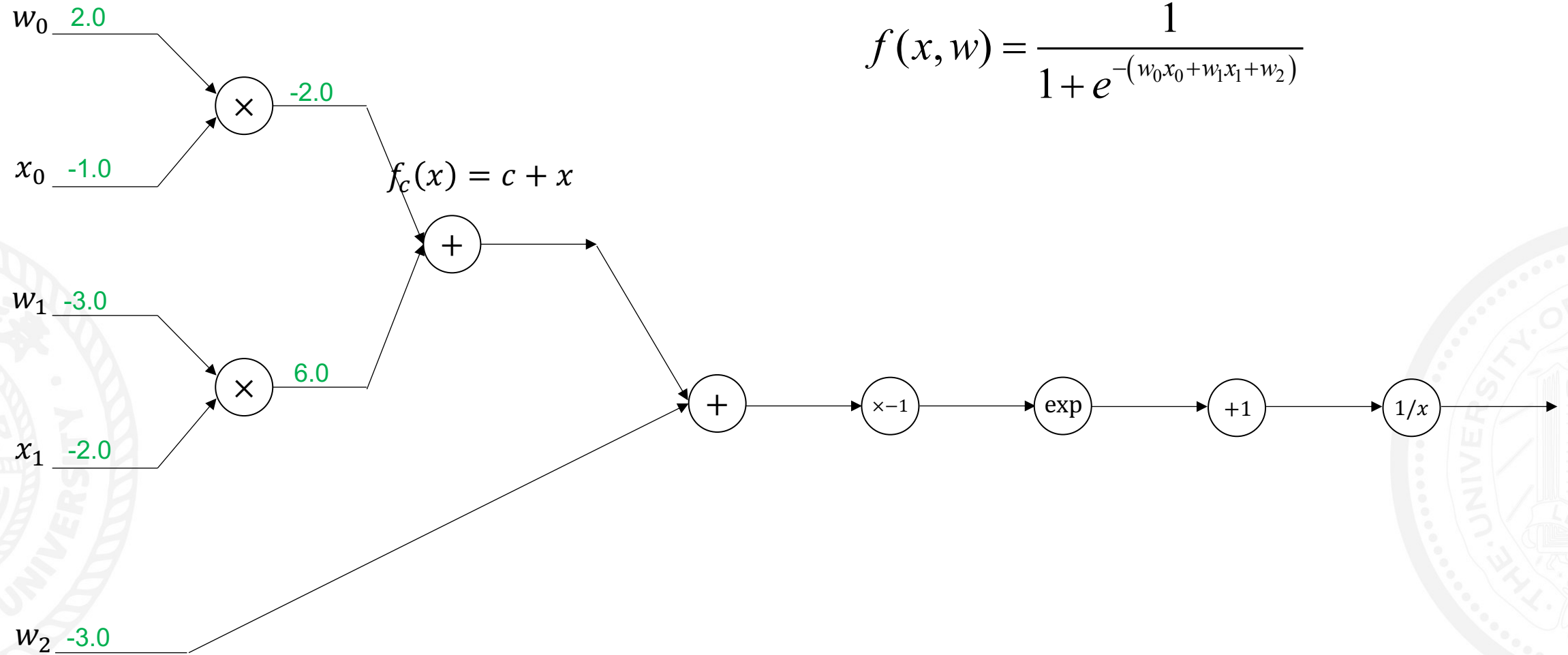


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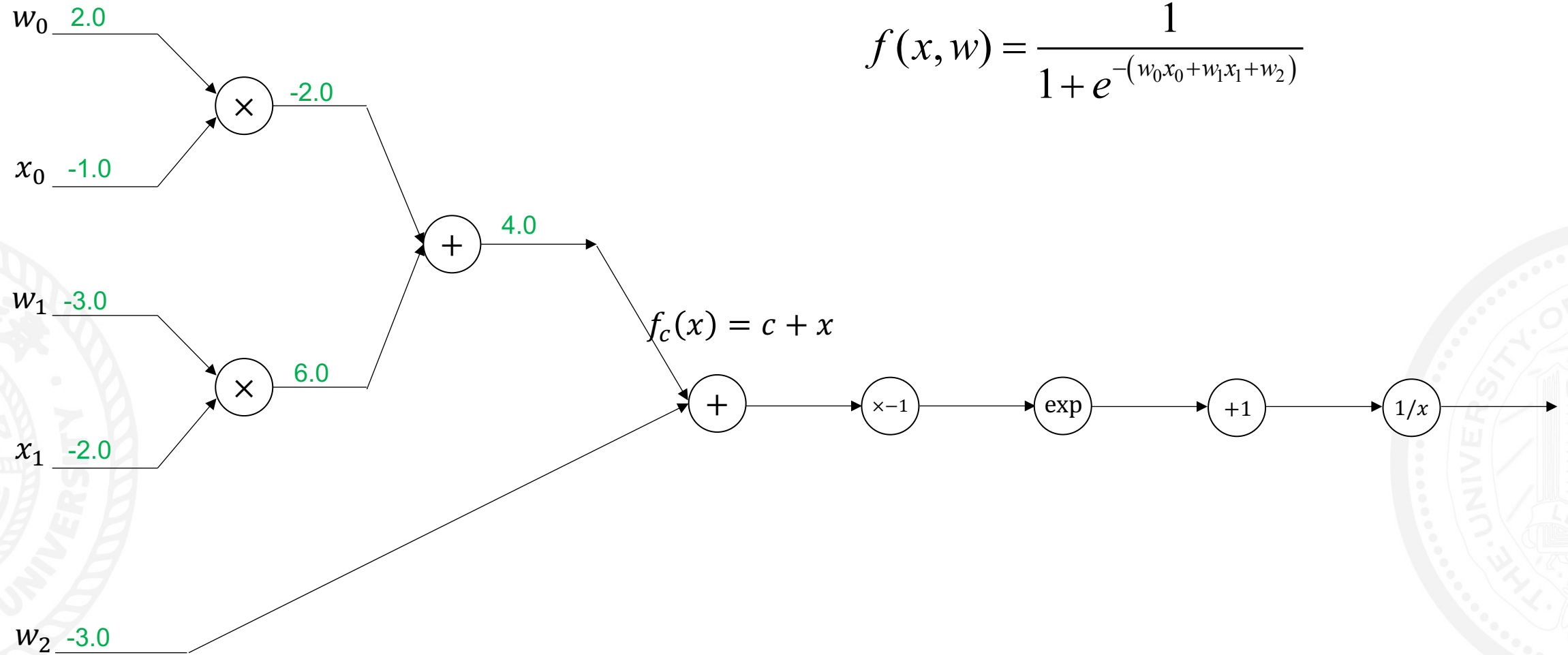


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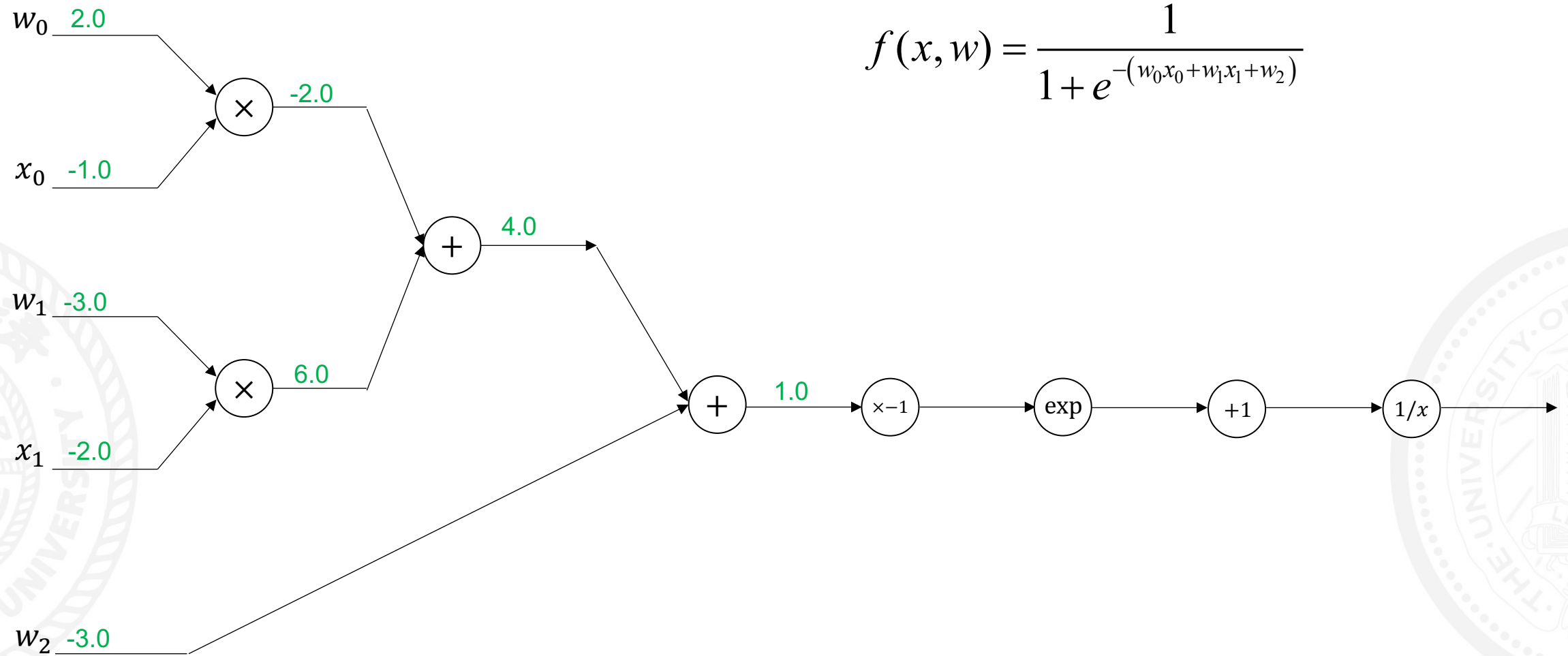


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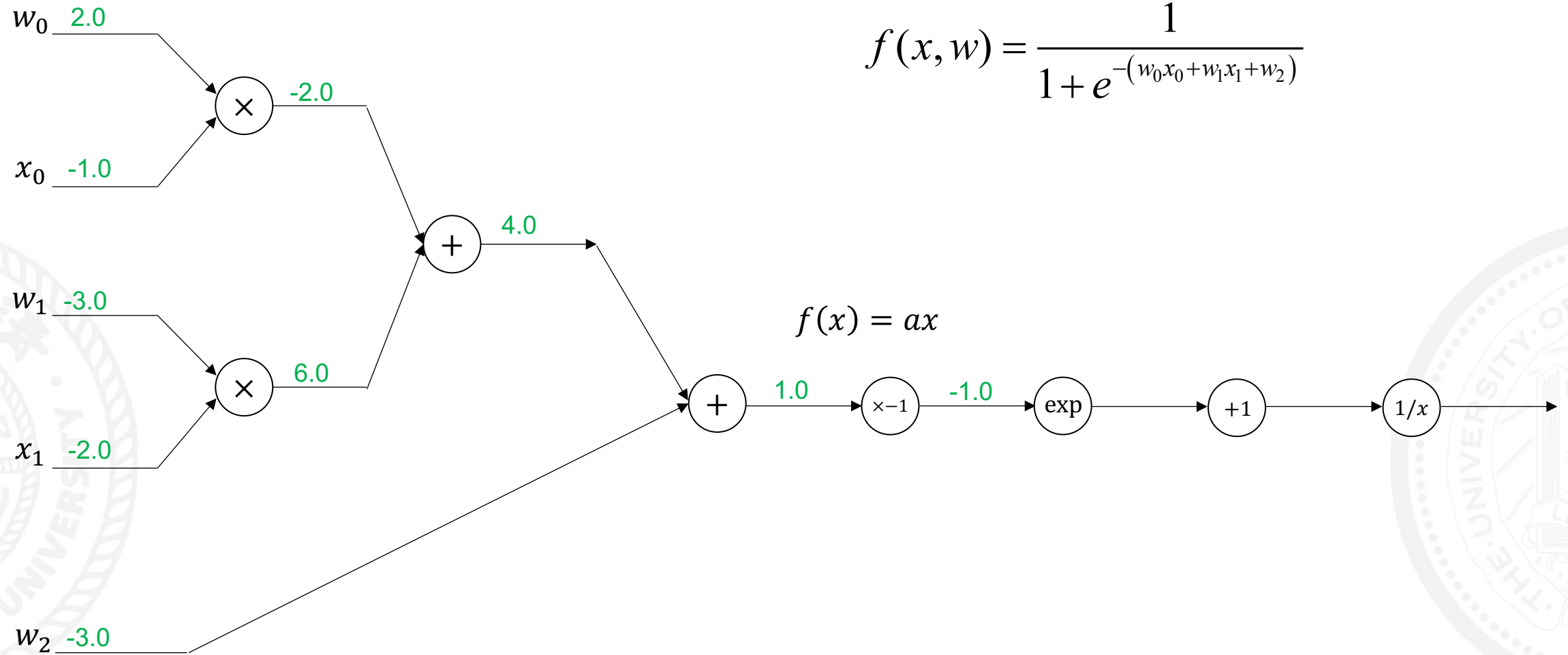


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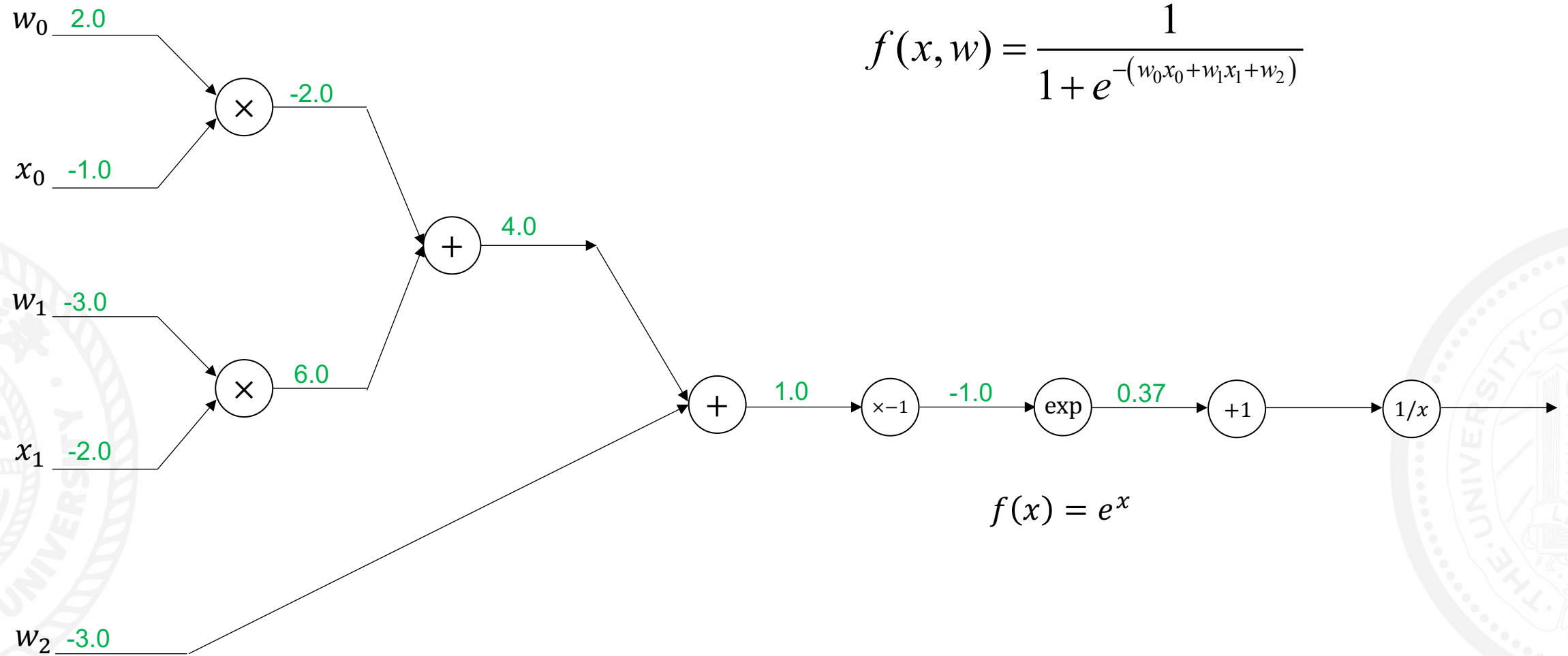


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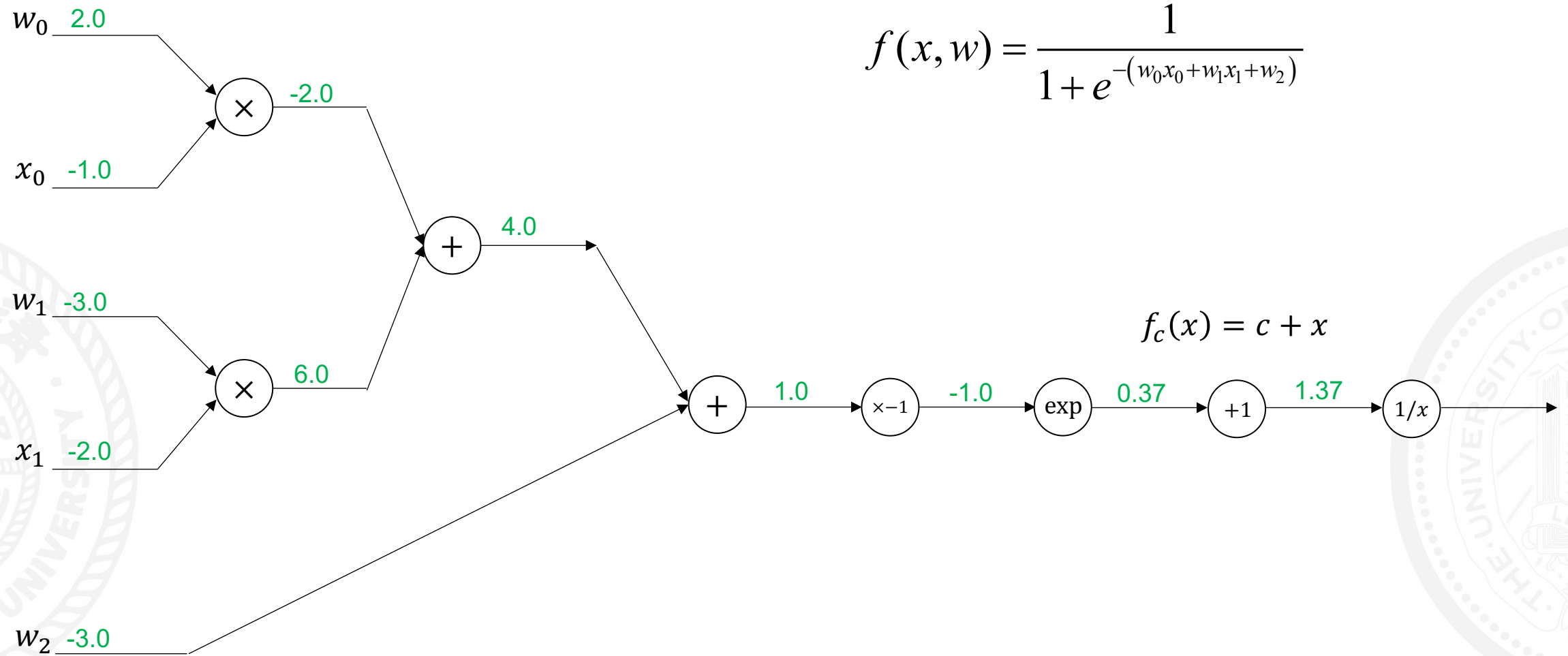


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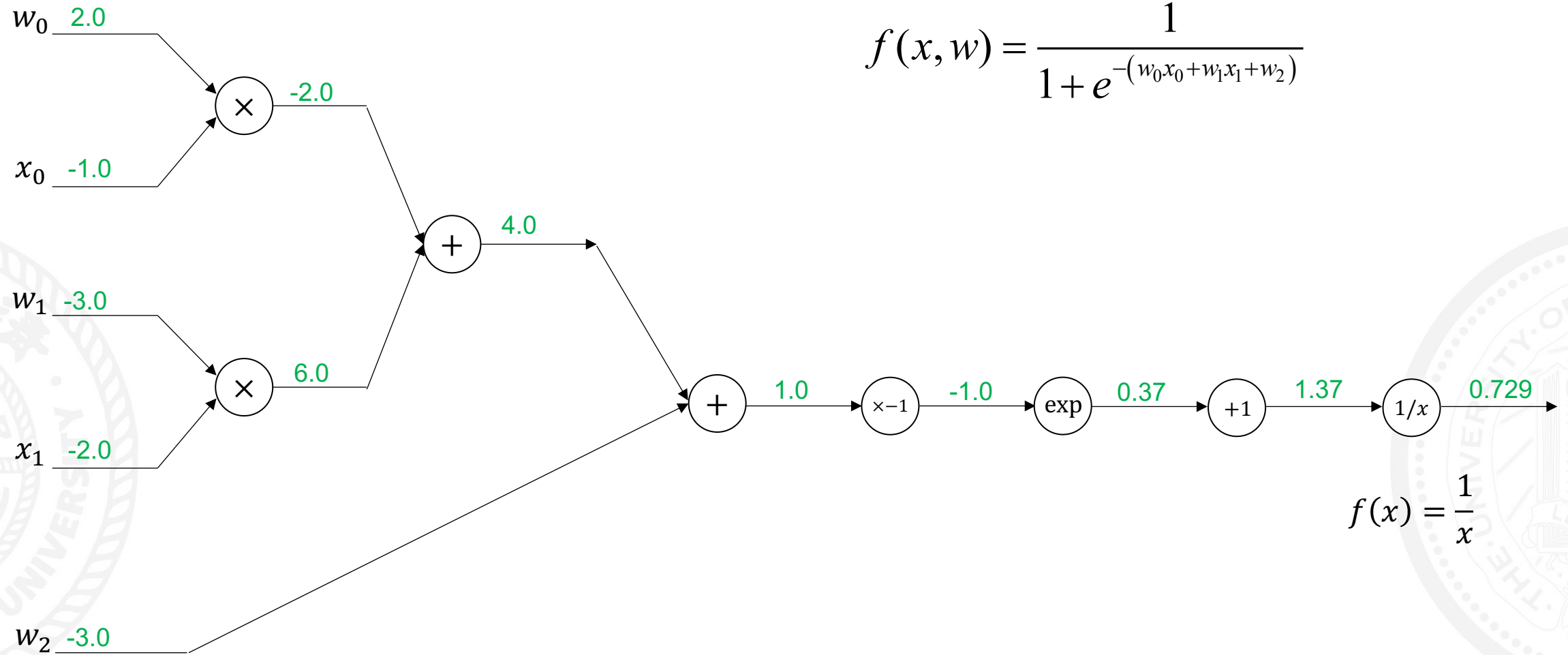


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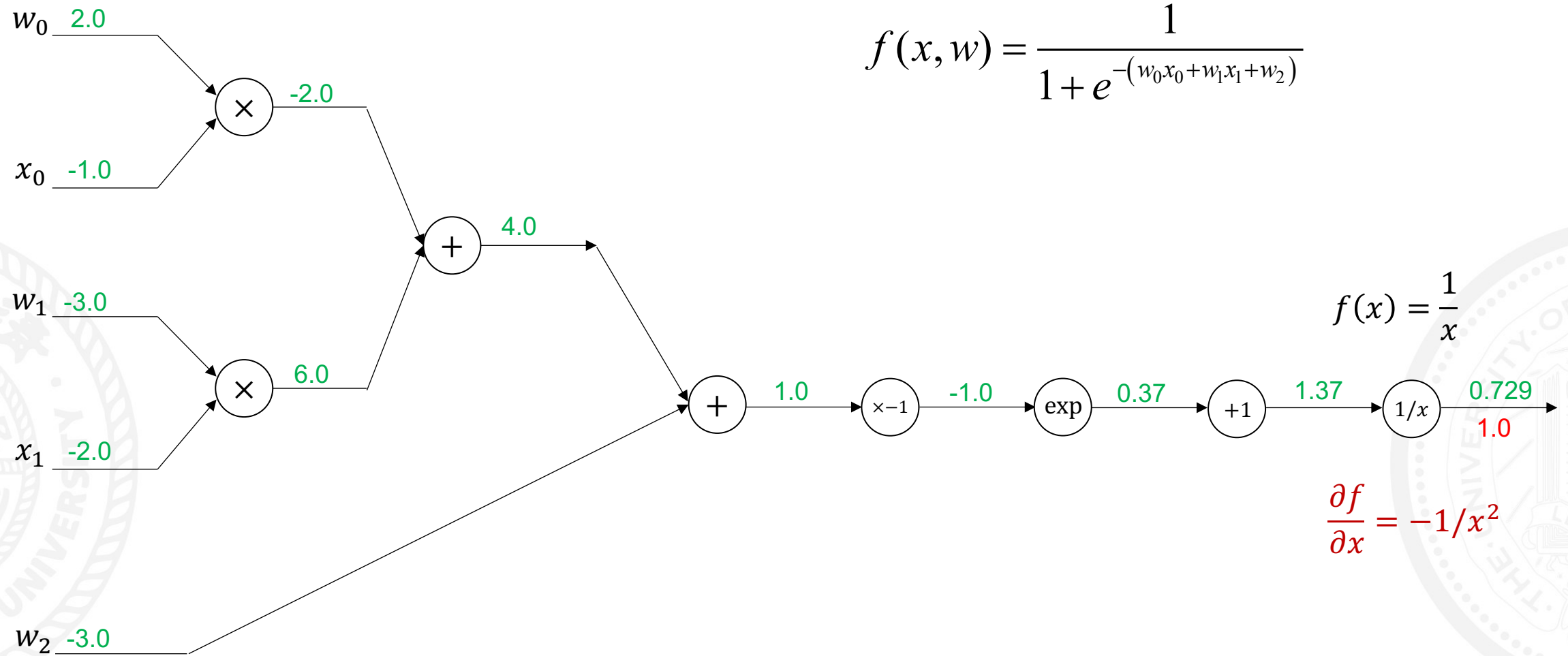


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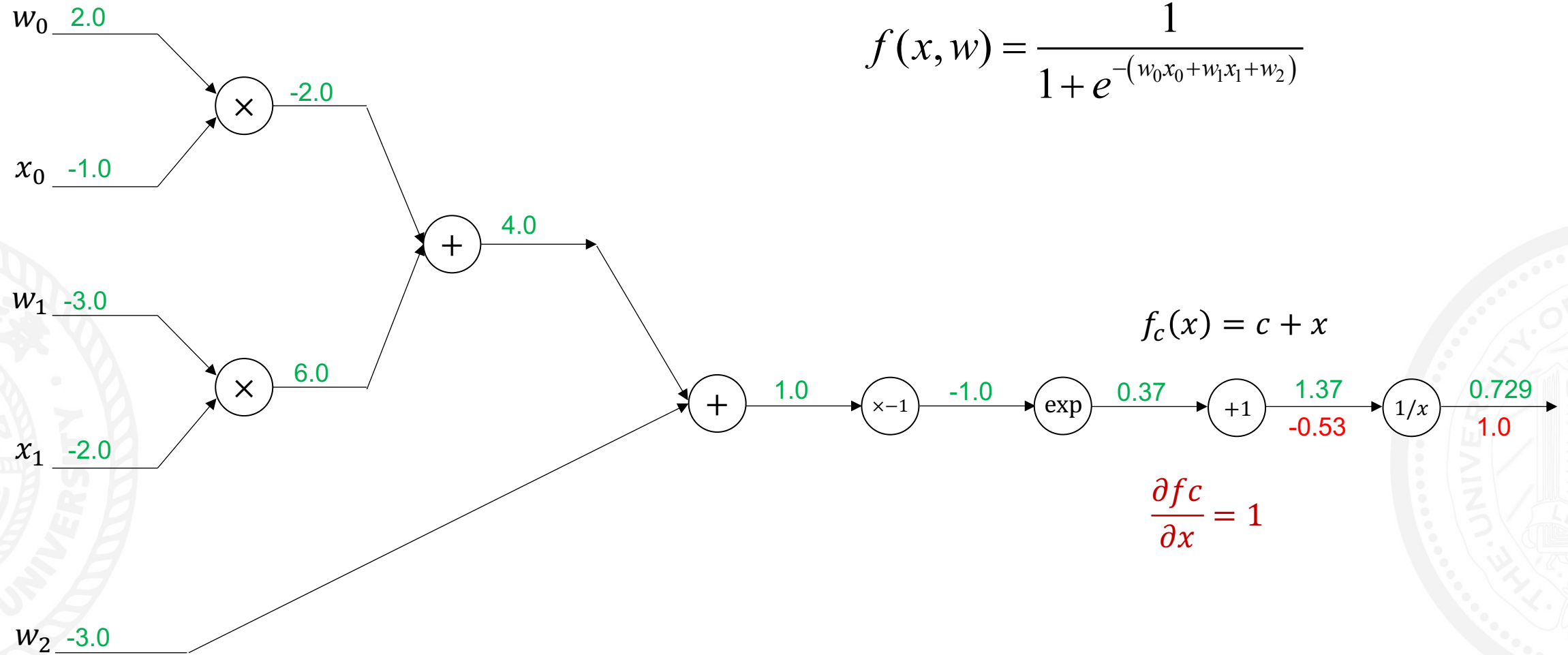


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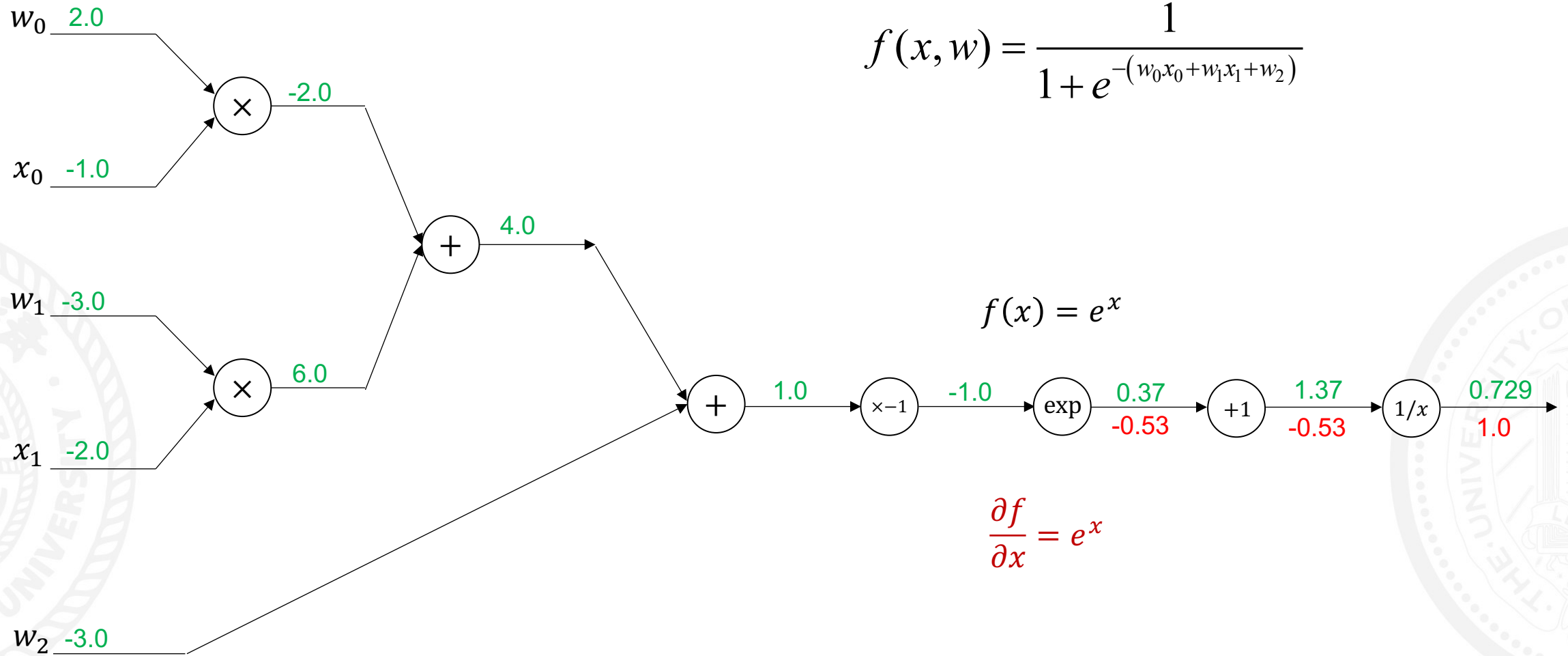


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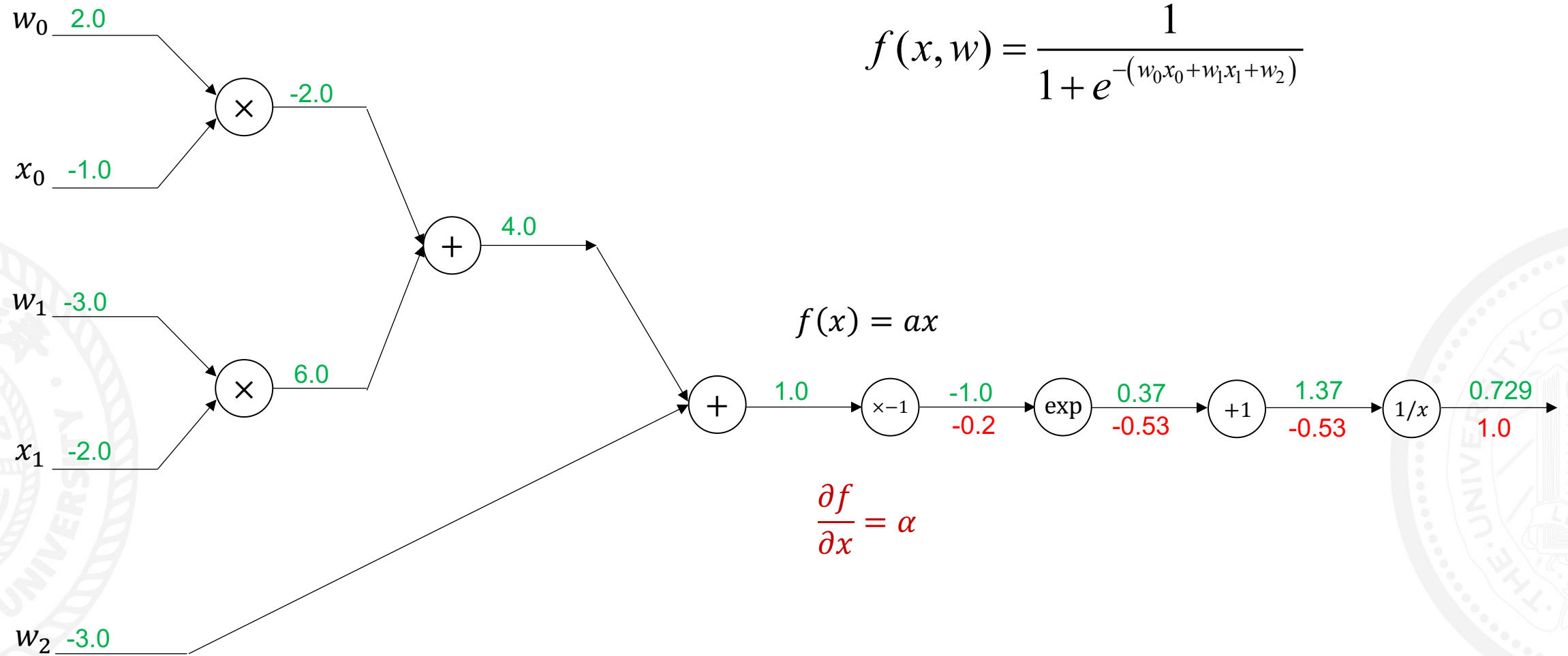


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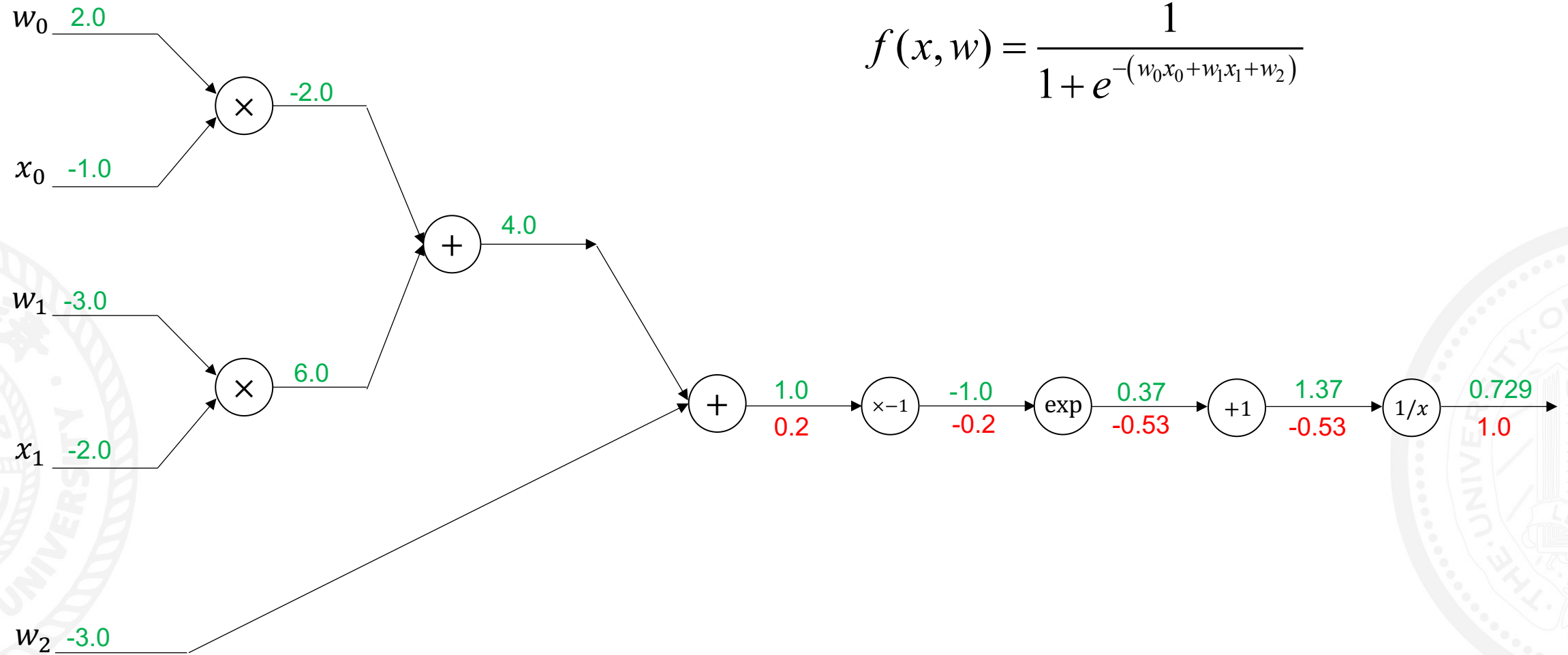


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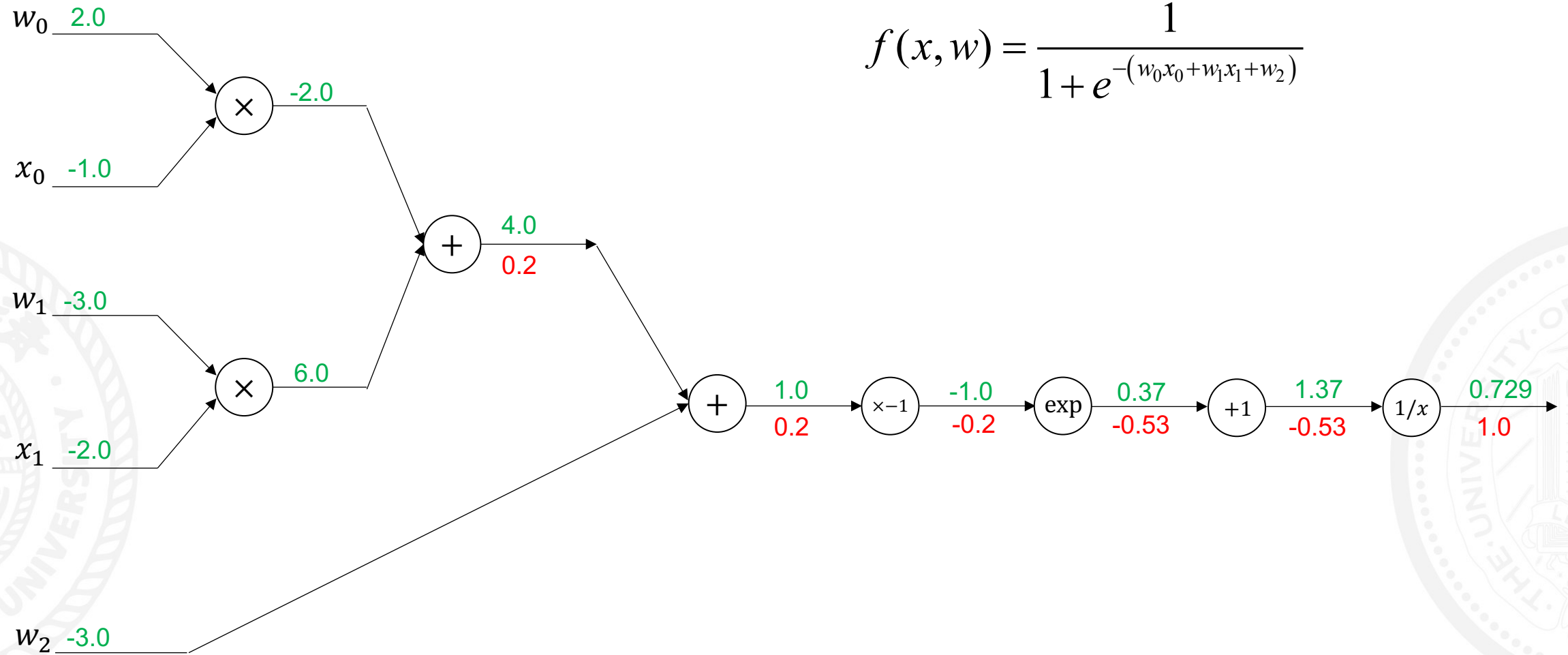


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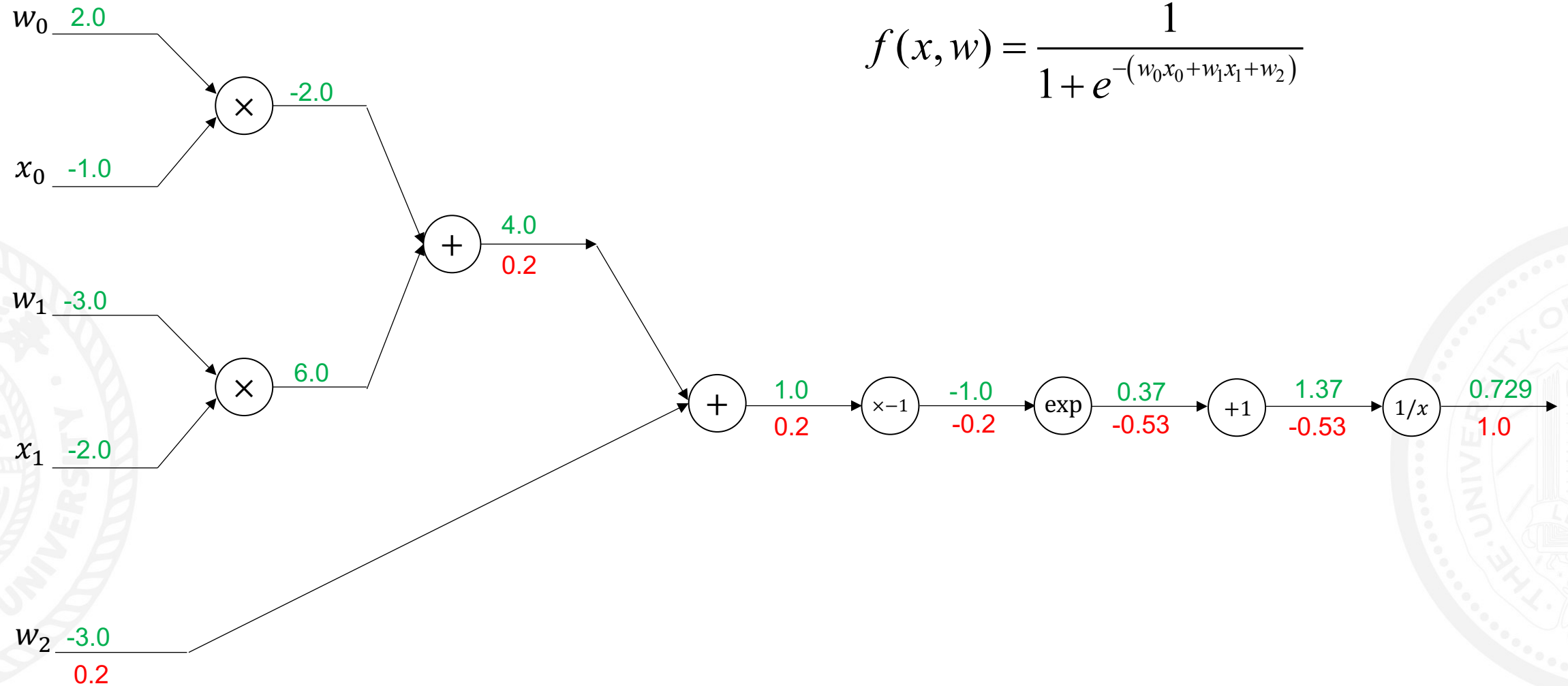


Backpropagation



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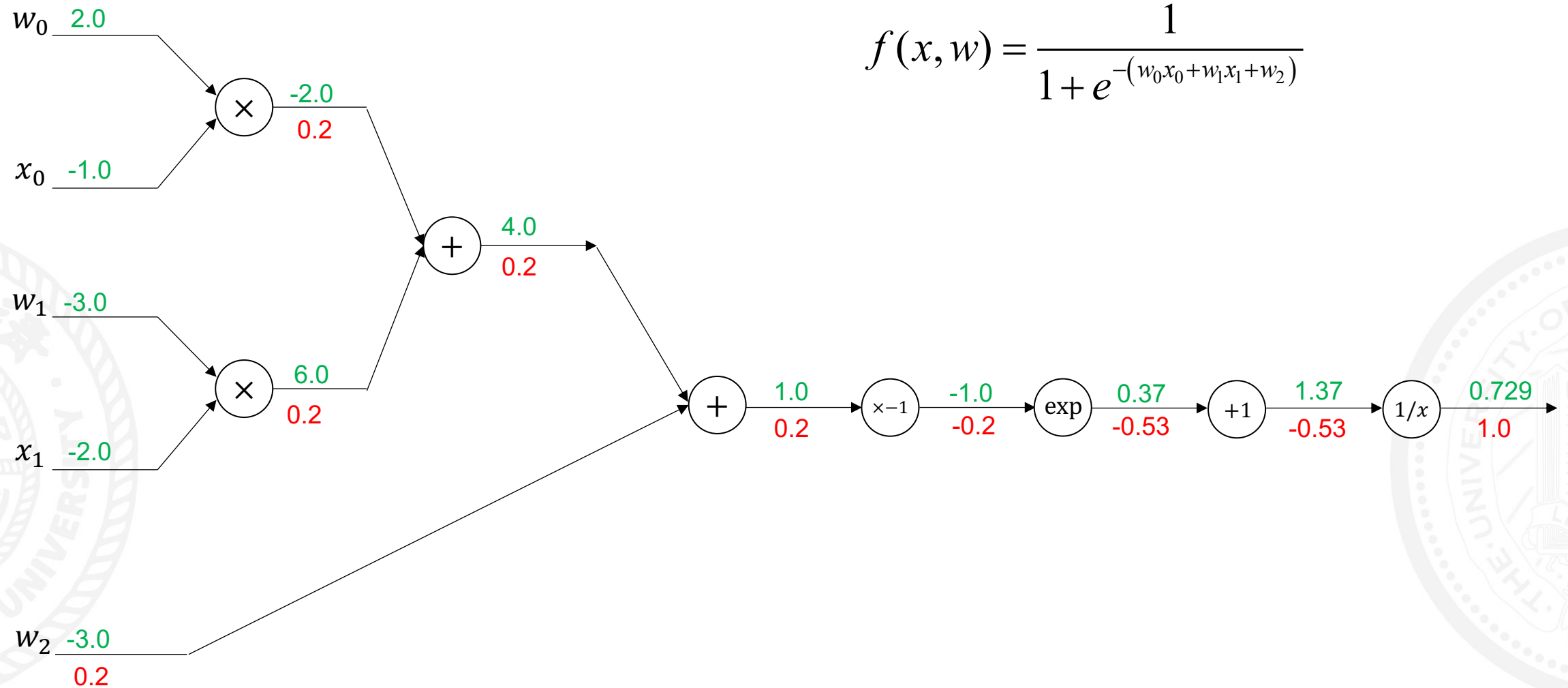


Backpropagation



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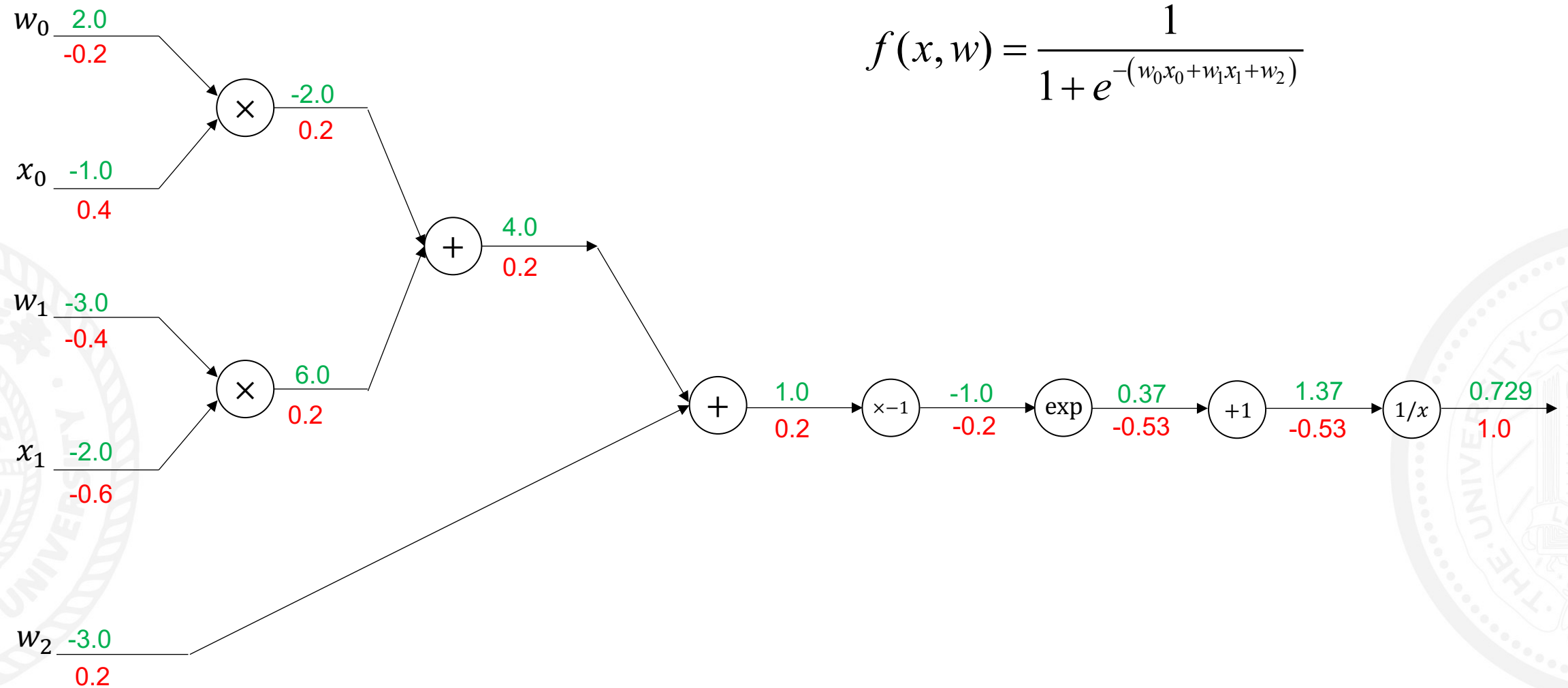


Backpropagation



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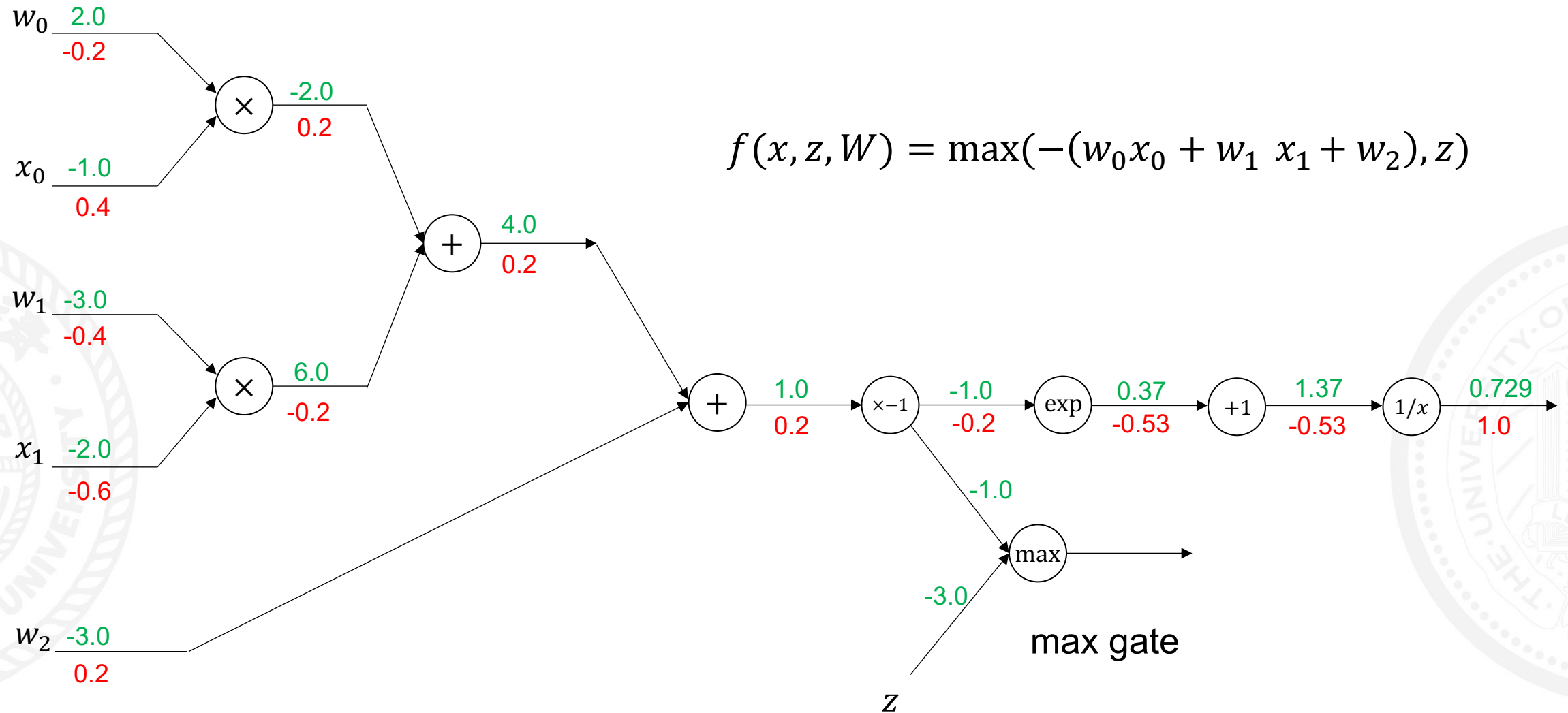


Backpropagation



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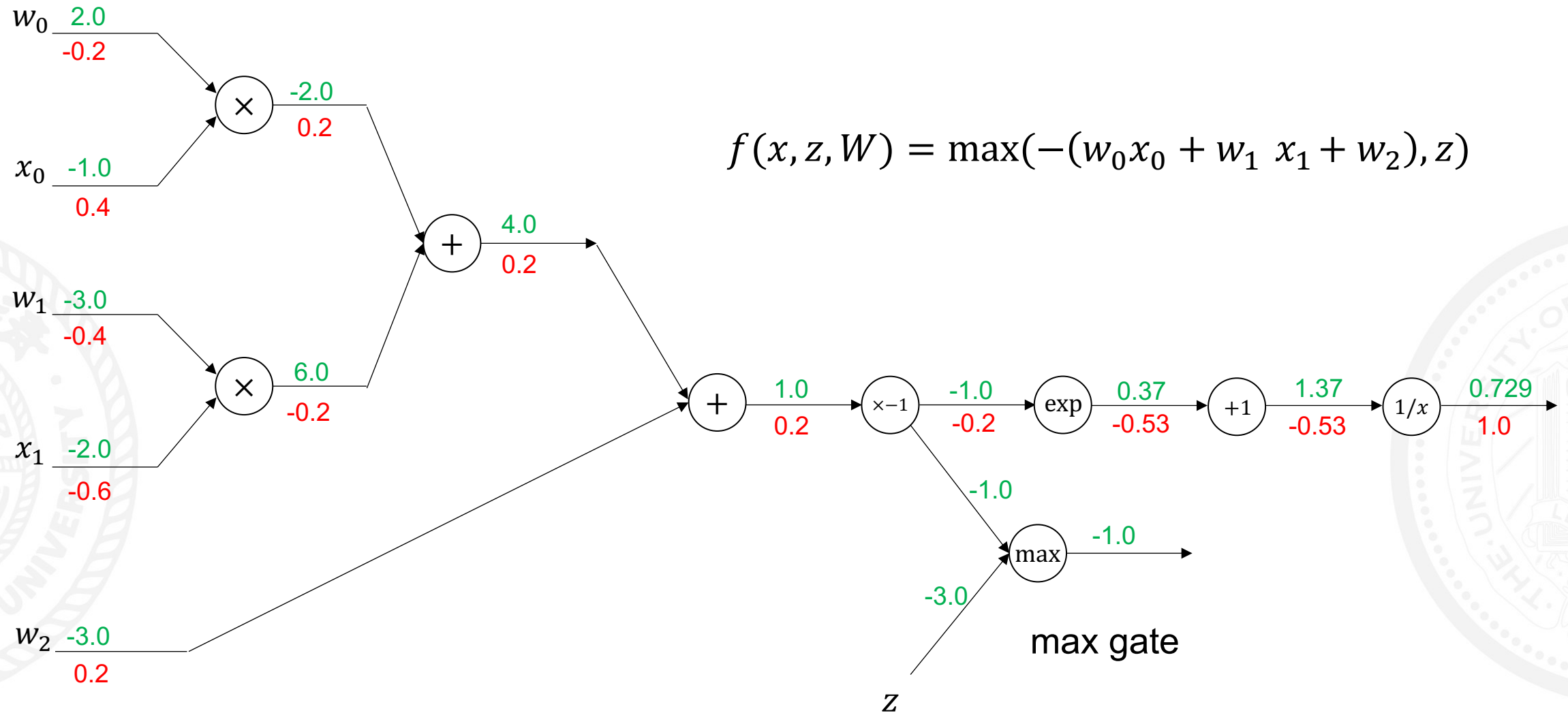


Backpropagation



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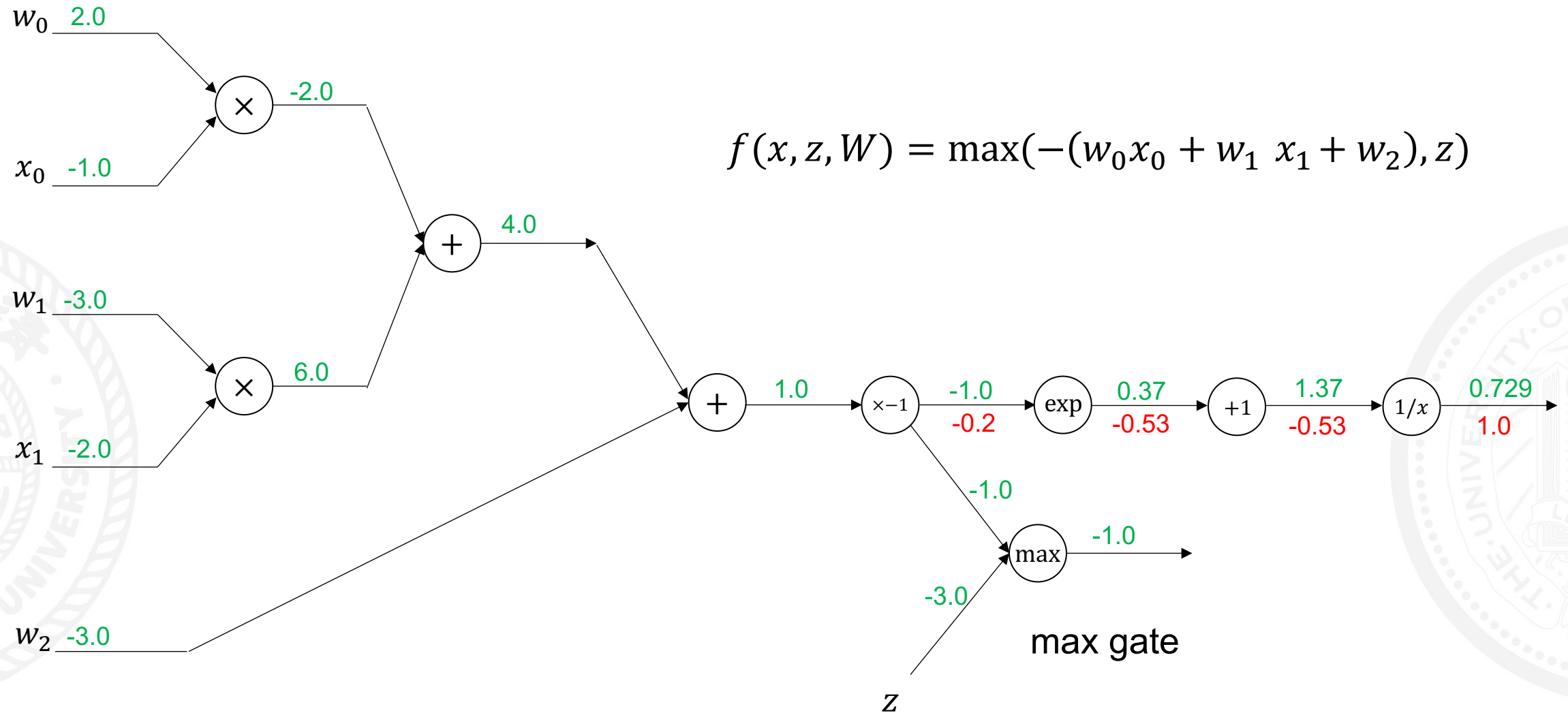


Backpropagation



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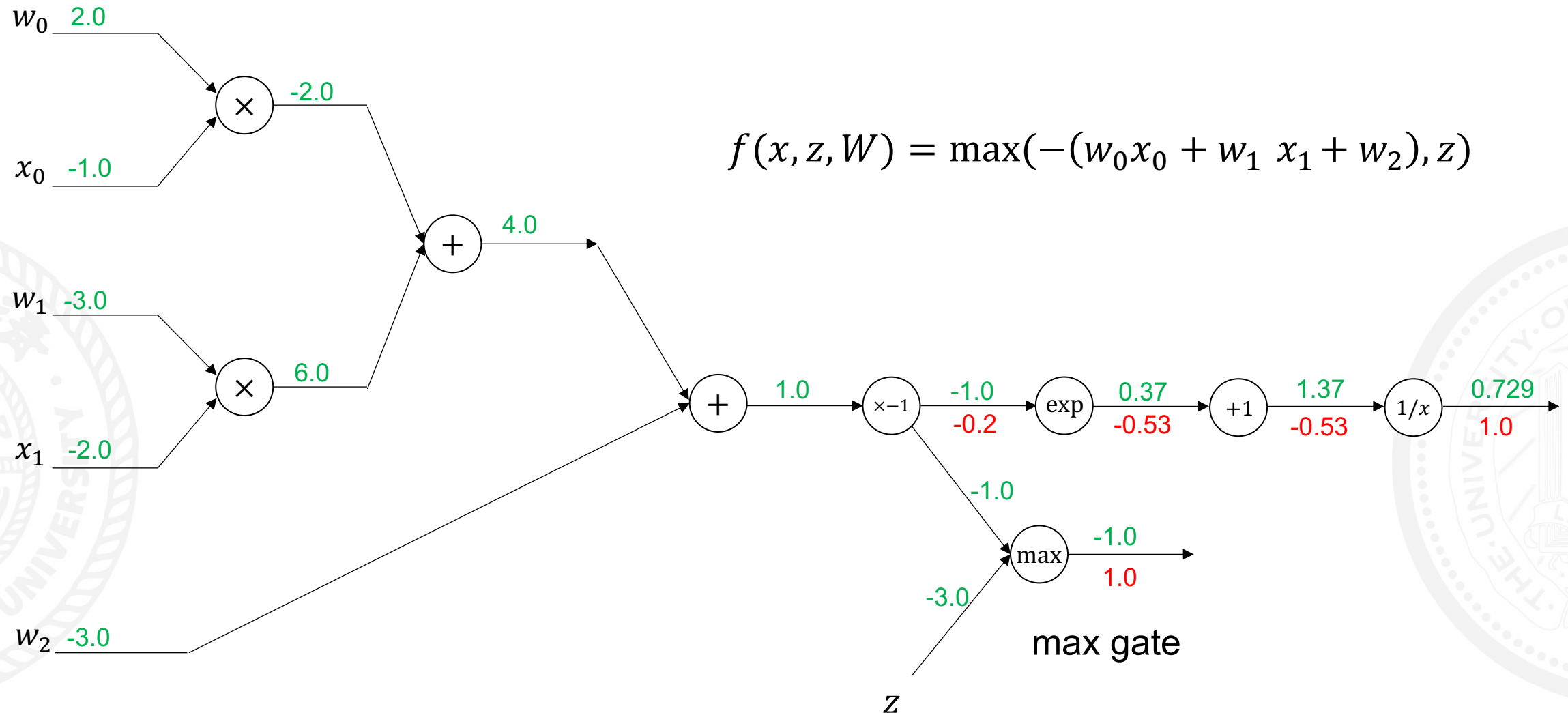


Backpropagation



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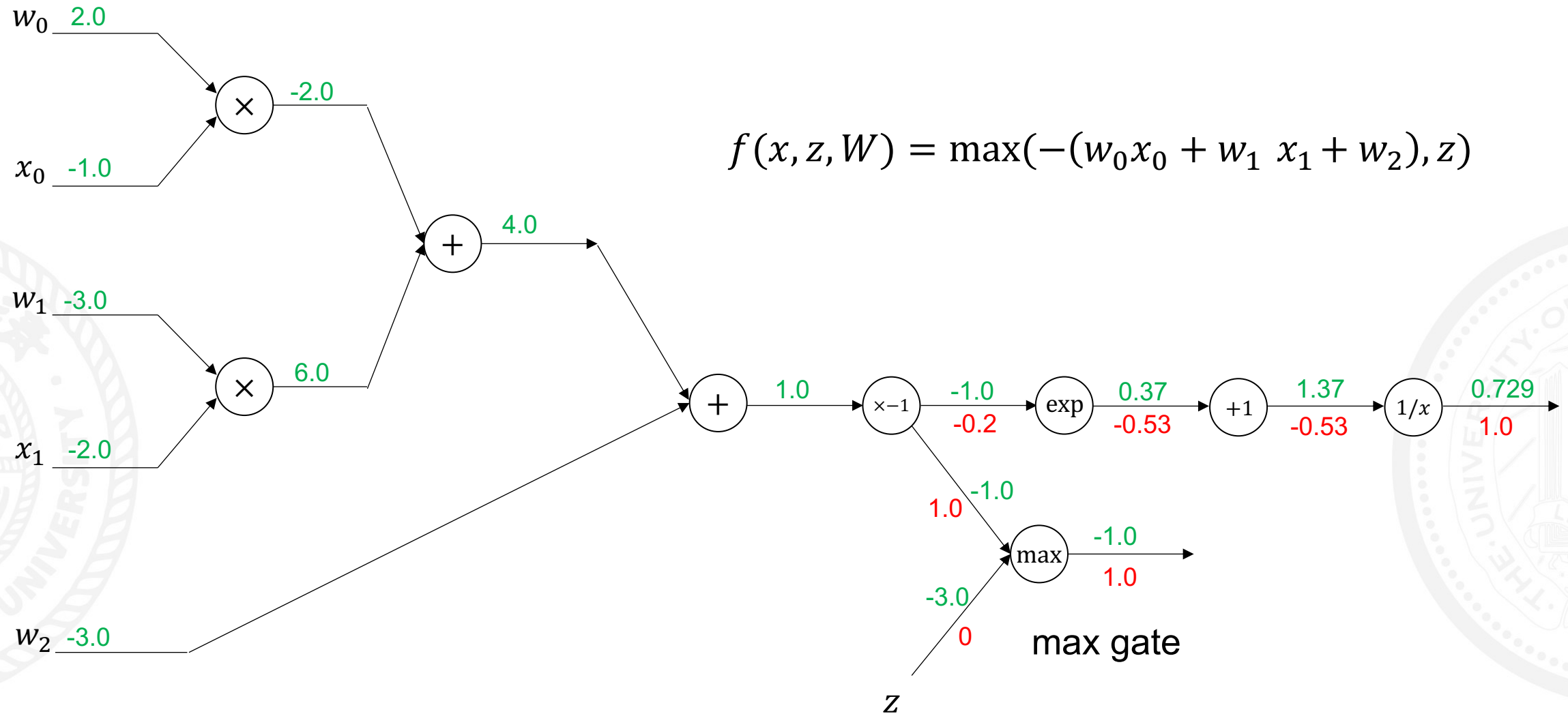


Backpropagation



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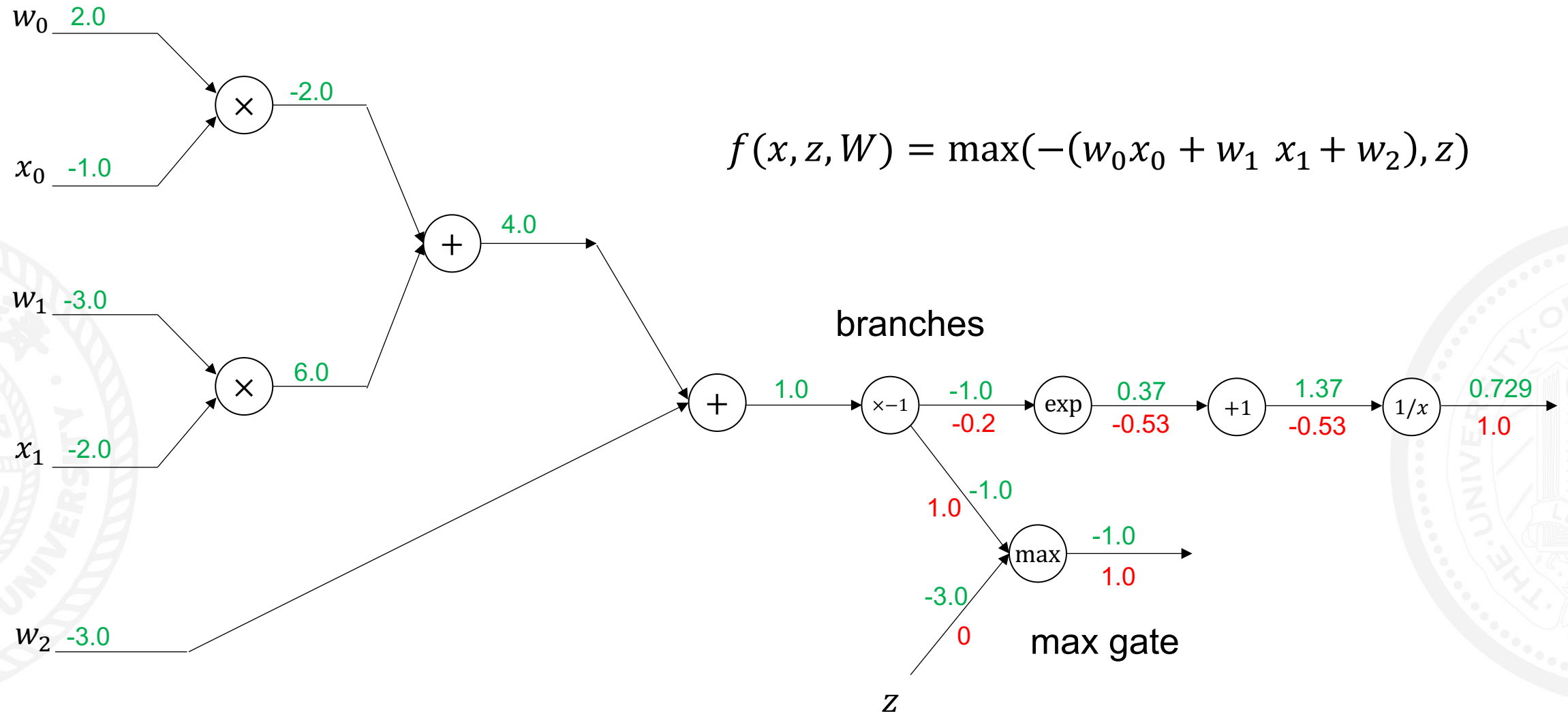


Backpropagation



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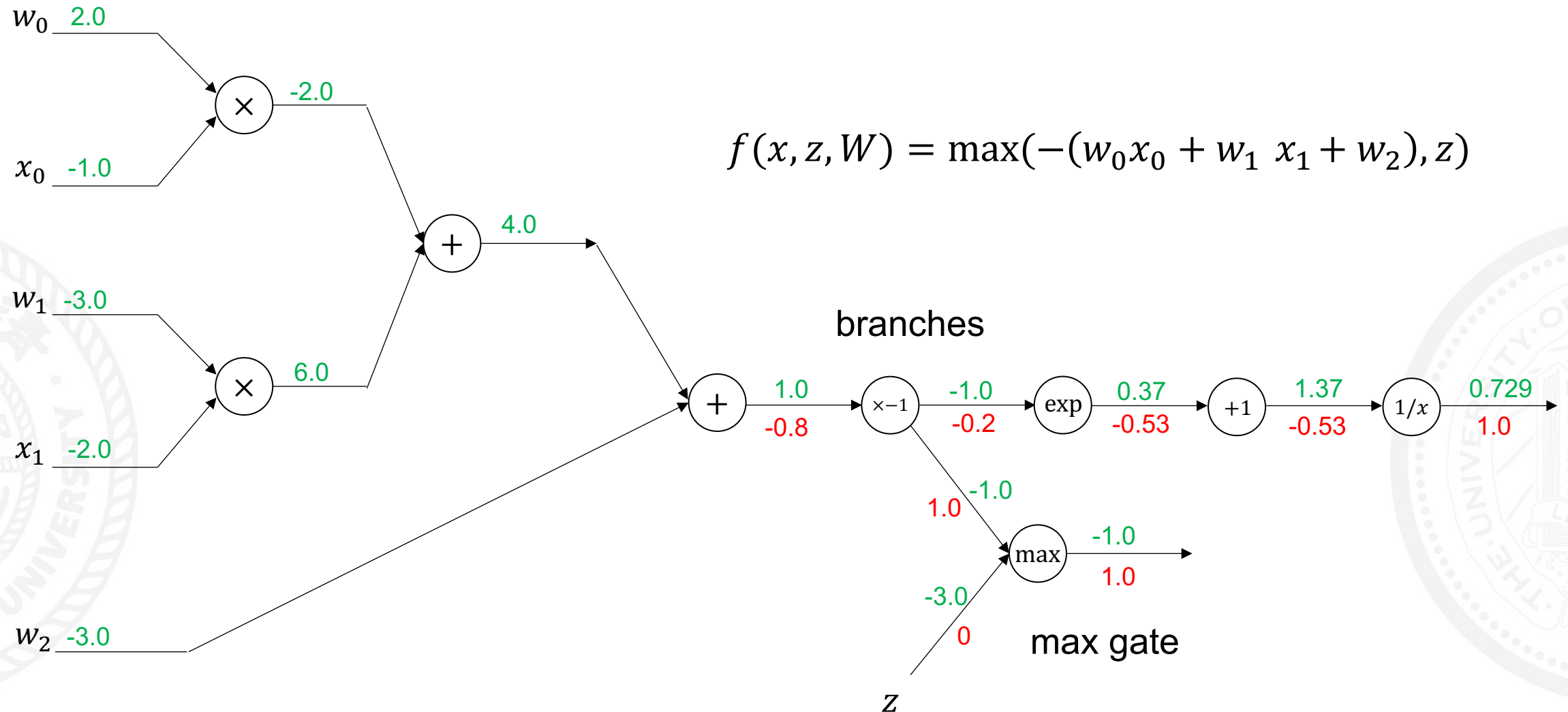


Backpropagation



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Backpropagation

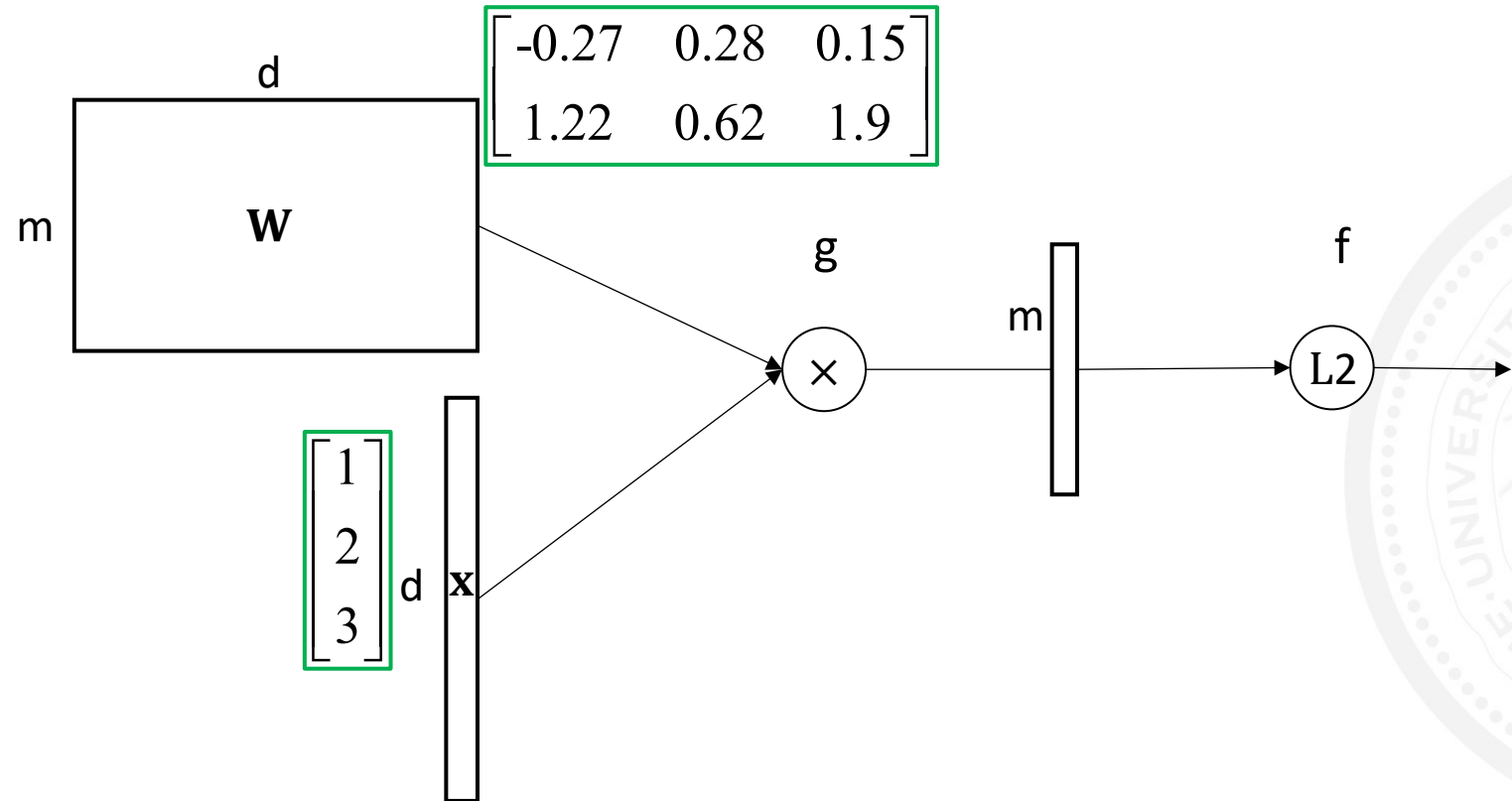


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Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$



Backpropagation

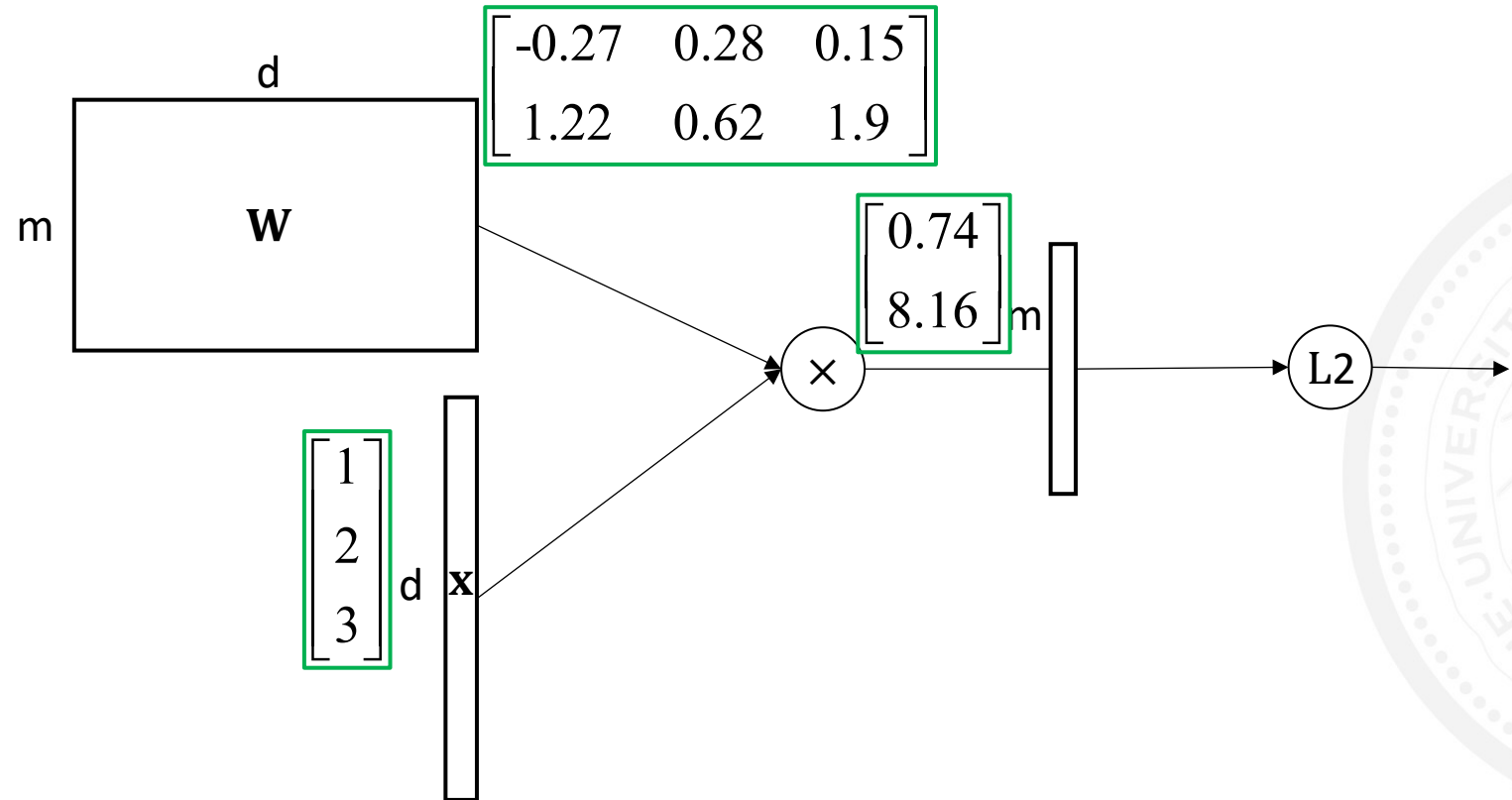


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Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$



Backpropagation

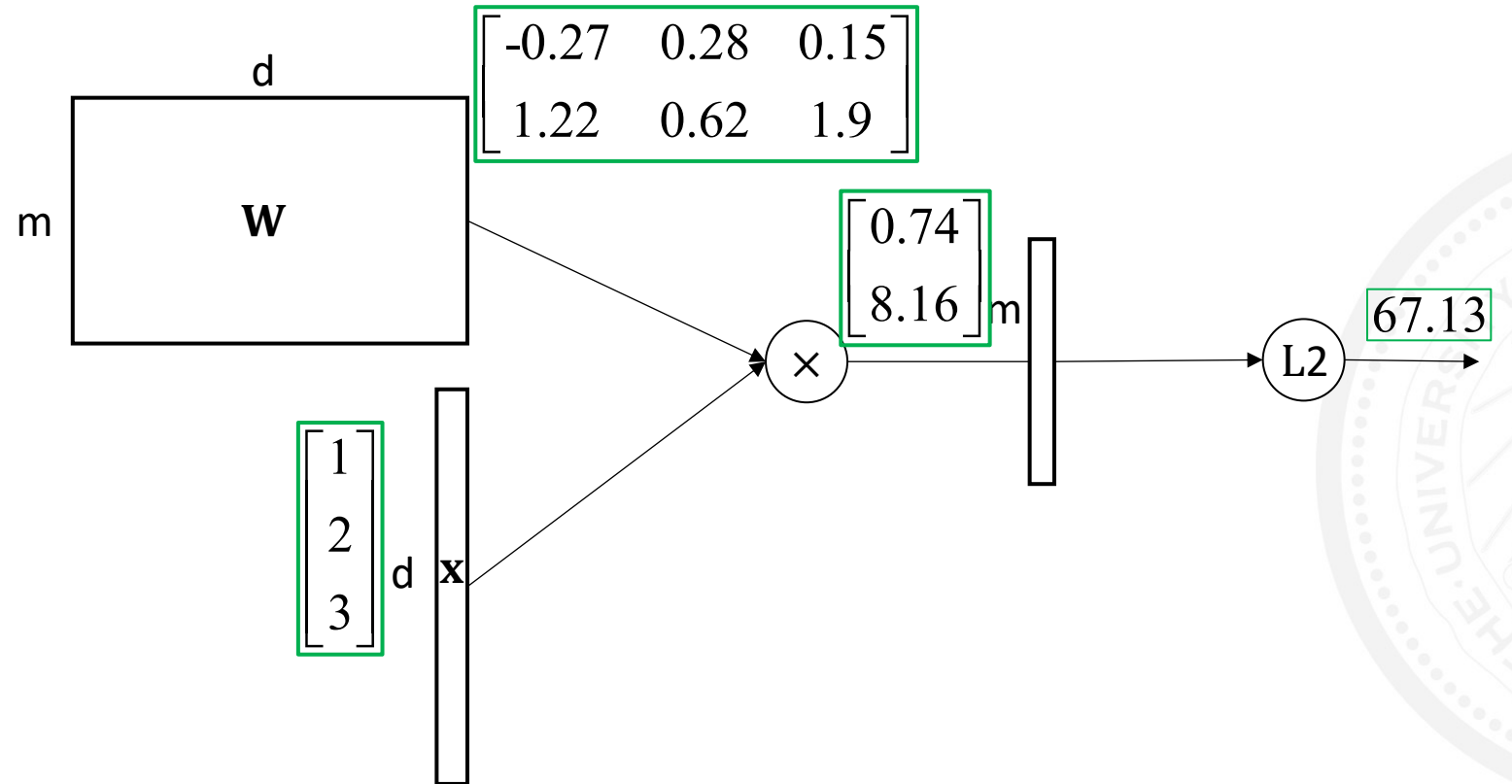


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Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$

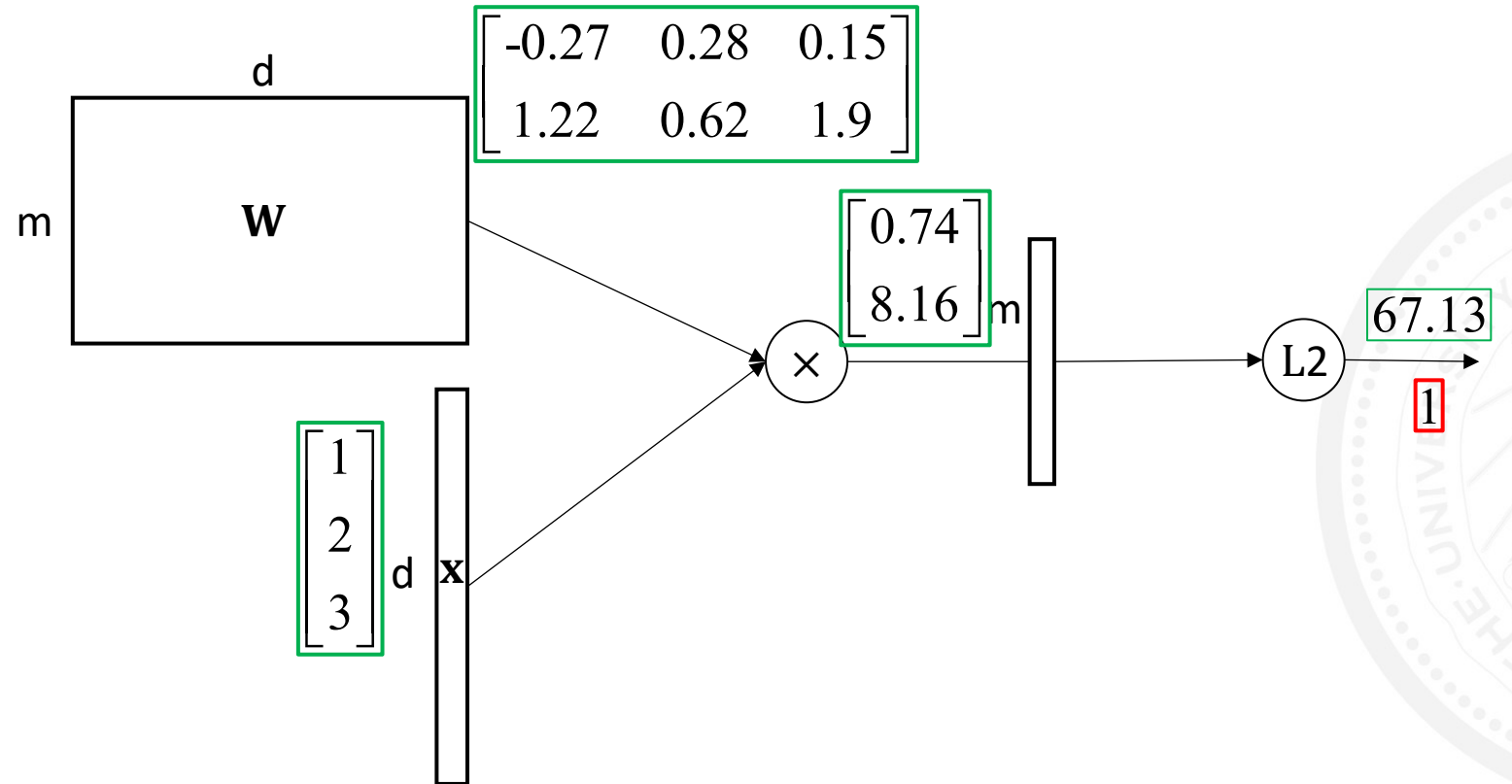


Backpropagation



Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$

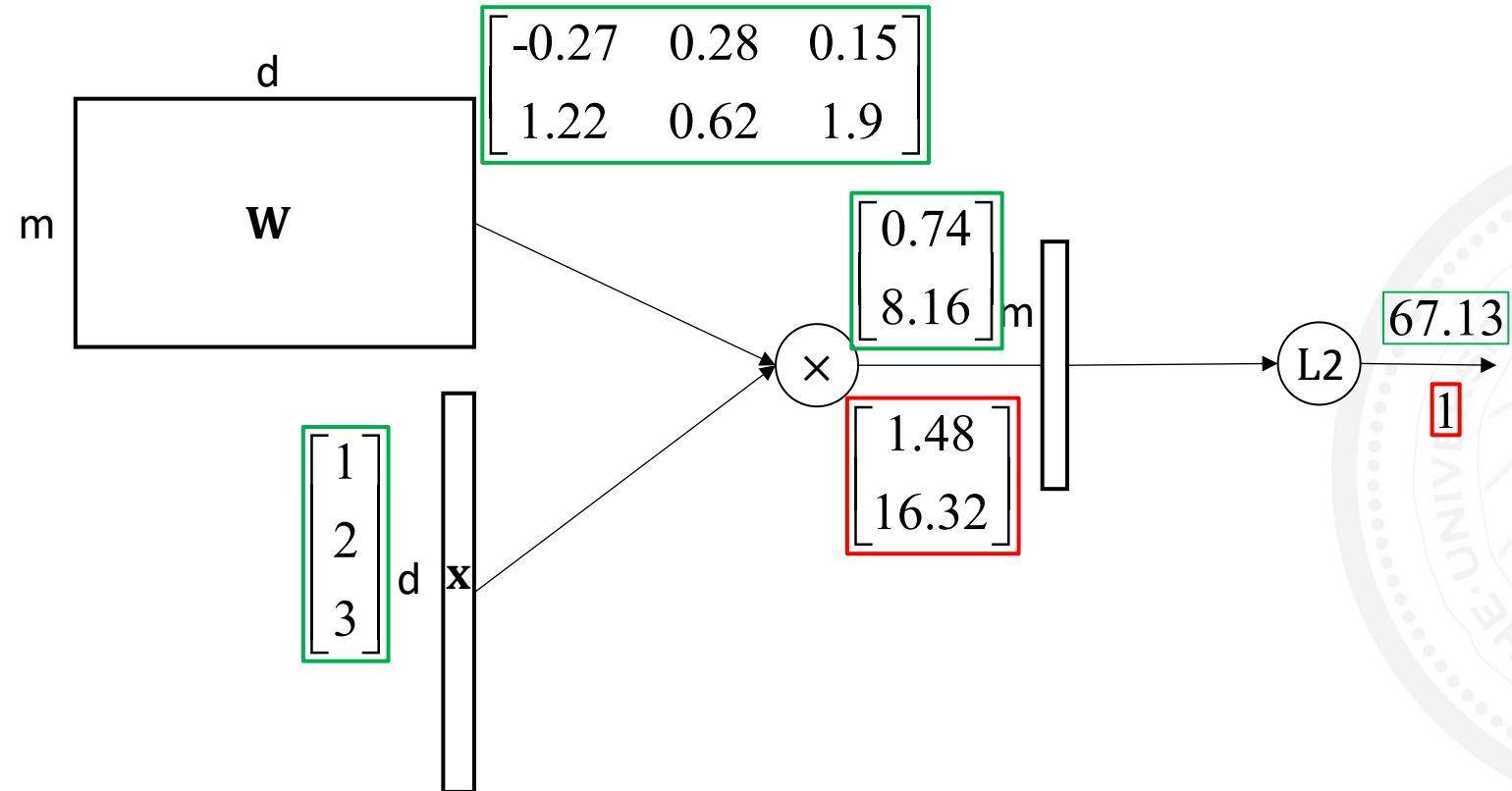


Backpropagation



Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$



Backpropagation



Vectorized example

$$f(\mathbf{x}, \mathbf{W}) = \|\mathbf{W} \cdot \mathbf{x}\|^2$$

